Evaluating the Potential of Zinc-ion Batteries

Abstract

As the world transitions to renewable sources of energy, massive grid energy storage will be necessary to accommodate the variable output of these sources. Grid solutions must be efficient, sustainable, affordable, and scalable to keep up with ever-increasing demand. While lithium-ion batteries are the current first choice for grid storage, zinc-ion batteries (ZIBs) are more sustainable and cost-effective. ZIBs have more abundant materials, lower prices, and less potential for harm. This study assesses the performance of zinc-ion batteries made using consumer-grade materials to determine the feasibility of these batteries on a commercial scale. Variations in the composition of the cathode and electrolyte were evaluated for improvements in the discharge profiles and overall performance.

Despite the heavy limitations placed on battery construction by the low-quality materials and simpler design, the batteries were able to effectively charge and discharge without a considerable loss in capacity. It was found that the zinc chloride used in later iterations was a more dangerous and unstable electrolyte, while the addition of manganese dioxide showed substantial improvement to battery performance in Iteration 5. Issues including short-circuiting and gas evolution while recharging demonstrated the need for improved design and construction. Nevertheless, the findings of this experiment underscore the usage zinc-ion batteries for grid storage while emphasizing the need for significant optimizations in the efficiency, stability, and longevity of zinc-ion batteries. Thus, this proof-of-concept study provides a foundation for further research into affordable and sustainable battery technologies.

Most renewable energy sources, such as wind and solar power, cannot continuously supply energy that is sufficient for the immense energy demands of human societies. Addressing this challenge will require advancements in technology, increased focus in sustainability, and robust grid storage solutions to ensure consistent energy delivery (Tang et al., 2019).

Of the current rechargeable battery technologies, the one that would be used for this energy storage would be the lithium-ion battery (LIB). However, lithium-ion batteries are expensive, flammable, and can be quite toxic. Furthermore, lithium supply is limited and not distributed equally around the world. Thus, certain countries may have political motives to control world lithium production, to the harm of others. Additionally, the unsustainable mining efforts required to mine lithium challenge the viability of lithium-ion batteries to help with the transition to renewable energy sources. Nickel-metal hydride (NiMH) batteries will not perform well in this role either due to the long time period over which the battery must store energy, as NiMH batteries lose considerable amounts of energy even while unused in storage (known as "self-discharge"). This effect renders NiMH batteries ineffective and inefficient for usage in grid storage technology (Silberberg, 2012).

In contrast, zinc-ion batteries (ZIBs) offer several advantages over the common lithium-ion batteries for grid storage. ZIBs are much less toxic, non-flammable, and can become much cheaper to produce. Zinc is over ten times as abundant as lithium and far easier to obtain (Van Noorden, 2014). Zinc is widely used in alkaline batteries due to its stability in non-flammable aqueous solutions, low cost, and large electrochemical potential.

Engineering Goal

This project was conducted to analyze and evaluate the feasibility of zinc-ion batteries using simplified designs and consumer-grade materials to evaluate the long-term viability of these batteries. To do so, the composition of the cathode and the electrolyte were varied to produce the results obtained in this paper. The different battery designs used in this experiment were intentionally left far simpler than those that would be used in building a similar battery at an industrial scale. This allows for a simpler investigation into the workings of the battery and simplifies the evaluation of battery performance.

History

Alessandro Volta created his first preliminary "battery" from the findings of fellow scientist Luigi Galvani in 1791. Volta's first effective battery, known as the voltaic pile, was invented in 1799 and detailed in a letter he sent to Sir Joseph Banks of The Royal Society of London (Volta, 1800). Pairs of copper and zinc discs were stacked on top of each other, each separated with brine-soaked cloth. While this battery was not very useful, it was the first demonstration of using chemical reactions to produce an electrical potential and led the way for other major inventions in battery technology. The dominant battery types used today include the lead-acid battery, the alkaline battery, the lithium-ion battery, and the nickel-metal hydride battery. In particular, the lithium-ion battery is the one of the most important components for the development of modern electronics (N. Zhang et al., 2020).

Theory and Principles

A battery is composed of three essential parts: an anode, a cathode, and an electrolyte. The anode is the negative terminal of a battery, while the cathode is the positive terminal. Thus, the current of a battery goes from the cathode to the anode during discharge. In reality, charge is carried by electrons, which flow in the opposite direction of the current because of their negative charge. Batteries are built so that those electrons cannot travel within the battery and thus must travel through an external circuit in order to reach the cathode. To complete the circuit, the ions within the battery must transfer charge from the anode to the cathode as well. Thus, a battery does not store any charge, but instead forces the movement of charge (in the form of electrons) through a wire to create a current, making a battery a "charge pump". Following this analogy, the potential difference produced by the battery is essentially the "pressure" applied by the "charge pump" on the electrons and ions. After factoring the movement of charge of ions inside the battery, the battery has neither stored nor released any charge, but instead moved the charges from one end of the battery to the other while doing useful work. Often, batteries contain a separator, a material preventing direct contact between the anode and cathode (that would cause a short circuit) while allowing dissolved ions to pass, as shown in Figure 1.

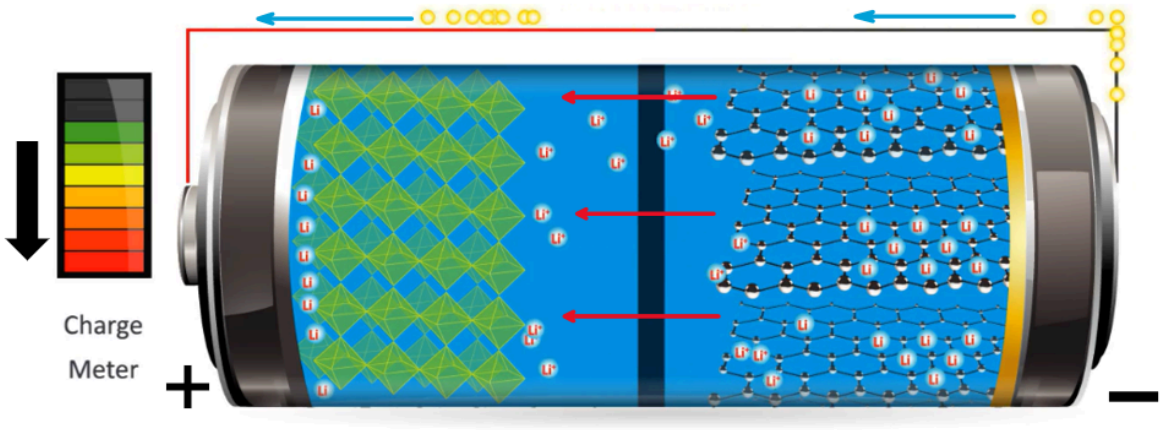


Figure 1: A model of the discharge of a lithium-ion battery. Electrons travel to the cathode following the blue arrows, while ions within the battery follow the red arrows through the black separator.

Almost all zinc-ion batteries involve the use of zinc metal for the anode. The anode reaction is represented by the following equation.

$$\mathrm{Zn}(s) \rightarrow \mathrm{Zn}^{2+}(aq) + 2e^{-} \tag{1}$$

The electrochemical potential of various reactions are defined using the standard reduction potential. Since a reduction cannot happen without an oxidation, the standard hydrogen reaction is assigned a reduction potential of 0 V, and all other standard reduction potentials are based on that (Cowie, 2014). The standard reduction potential is measured at a temperature of 298.15 K (25° C) and 1 M concentration solutions. The above reaction is the oxidation of zinc, which is simply the reverse of the zinc reduction reaction.

$$\mathrm{Zn}^{2+}(aq) + 2e^{-} \rightarrow \mathrm{Zn}(s) \tag{2}$$

The standard oxidation potential is the negative of the standard reduction potential. As Equation 2 is $E_{\text{reduction}}{}^{\circ} = -0.76\,\mathrm{V}$, the reaction of Equation 1 creates an electrochemical potential of 0.76 V.

As zinc is converted into zinc ions, it dissolves into the electrolyte. This electrolyte consists of a zinc compound dissolved in water, such as zinc sulfate or zinc chloride. A higher electrolyte concentration increases the solution conductivity, which is equivalent to decreasing the internal resistance of the battery. Decreasing the internal resistance allows for a higher maximum current from the battery. However, once the concentration is too high, the increase in viscosity due to the solute impedes the movement of ions in

the solution, increasing the internal resistance. Thus, finding the best electrolyte concentration to minimize the internal resistance of the battery is an important part of battery design. Different electrolytes also have different solubilities and conductivities, so varying the electrolyte can also help to minimize the internal resistance.

The cathode was constructed using a solid piece of graphite, graphite powder, and activated carbon powder, each of which serve different functions. The solid piece of graphite functions as a current collector, to which wires can be attached and through which a circuit can be completed. While graphite and activated carbon are both carbon allotropes, they serve different purposes here.

The purpose of graphite powder was to intercalate zinc ions. Intercalation is the insertion of a molecule or ion (the guest species) between the sheets of a layered compound (the host). Layered compounds are best described as a series of stacked sheets, with strong intermolecular bonding within the layer, but weaker bonding between layers. Graphite is the most well-known example of this phenomenon. As shown in Figure 2, a guest species can expand the layers of a layered structure so that the species can be stored between those layers (Inaba, 2009). The energy needed to expand these layers is normally derived from a redox reaction, such as the oxidation of zinc in this case. If the guest species is ionic, the host-guest combination will have a net charge. Intercalation is used in most "-ion" batteries, such as lithium-ion and zinc-ion batteries. For intercalation to happen in a zinc-ion battery, the host material should be the cathode, and ions in solution will intercalate into the layer structure while discharging.

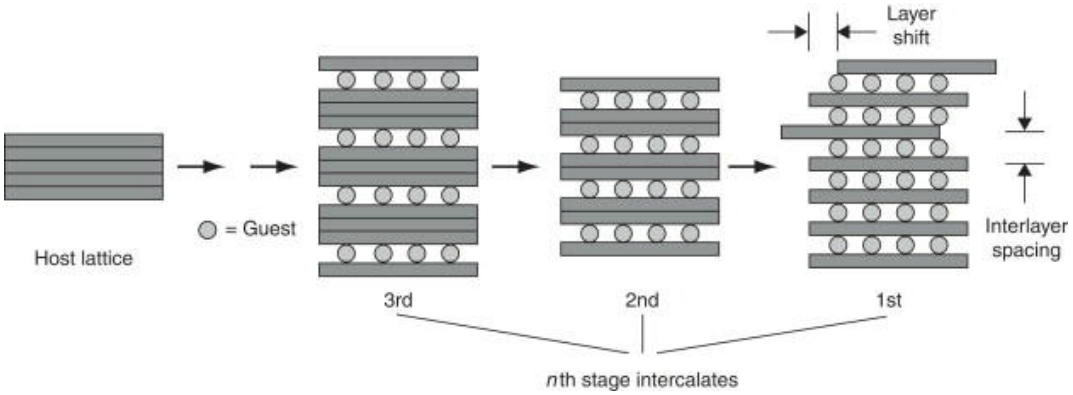


Figure 2: The main transitions during the intercalation of a guest species into a graphite host.

While intercalation and other possible cathode reactions are well-studied in lithium-ion batteries, there is less known about what occurs in zinc-ion batteries with a graphite cathode. Nevertheless, the

cathode reaction between zinc and the layers of graphite, as shown in Equation 3, contributes to the potential difference of the battery.

$$\mathrm{C}_x + \mathrm{Zn}^{2+} + 2e^- \rightarrow \mathrm{ZnC}_x \tag{3}$$

All graphite intercalation compounds have the formula XC_n, where $n \geq 6$ (IUPAC, 1994). However, the properties of zinc ions cause those ions to intercalate non-stoichiometrically, and can vary heavily depending on the method and the compound used for intercalation. Zinc ions are divalent, which allows for more complex bonding and interactions between the graphite and zinc. The distance between successive layers of graphite is about 3.5 Å on average (Çakmak & Öztürk, 2019). While zinc only has an ionic radius of 0.74 Å, anions (chlorine ion, sulfate ion, etc.) also intercalate between the layers. The anions have much larger ionic radii, such as 1.81 Å for chlorine ions and 2.42 Å for sulfate ions on average (Health, 2004). Besides those factors, electrostatic repulsion and disruption of the graphite structure due to the presence of these ions cause graphite to expand during intercalation.

The final iteration of the battery also included manganese dioxide (MnO_2) in the construction of the cathode. Manganese dioxide is another intercalating compound and has been found in some studies to be more or as effective as graphite for intercalation in zinc-ion batteries (Ming et al., 2019). Zinc and manganese dioxide are fundamental components to alkaline batteries, but a different reaction occurs in alkaline batteries which is not reversible. The reaction that occurs in zinc-ion batteries (which is reversible) is modeled by the following chemical equation:

$$\mathrm{Zn}^{2+} + 2\mathrm{MnO}_2 + 2e^- \longrightarrow \mathrm{ZnMn}_2\mathrm{O}_4 \tag{4}$$

The final part of the cathode is activated carbon powder, which is also known as activated charcoal. Activated carbon has an extremely high surface area of over $3\,000\,\mathrm{m}^2\,\mathrm{g}^{-1}$. This makes activated carbon useful for filtration, odor removal, and nullifying toxins, among other uses (Islam et al., 2021). Activated carbon does not have the layered structure necessary for intercalation, but the extremely high surface area allows for adsorption of zinc ions onto the surface of the powder. Adsorption is a process by which one material, the adsorbate, adheres to the surface of another material (the adsorbent) (López et al., 2023).

Activated carbon is able to adsorb zinc ions to its surface through the use of van der Waals and London dispersion forces. Van der Waals forces can be caused by a variety of factors, but the most likely one here is the Debye force, where momentary dipoles that form and disappear in the graphite attract the positively charged zinc ions. London dispersion forces arise in the interaction between momentary dipoles in both molecules. However, since these zinc ions are positively charged, an outer layer of negatively charged ions forms loosely around the inner layer of positive zinc ions. This phenomenon is known as the electric double layer (EDL), and is visible in activated carbon due to the enormous surface area. However, studies have shown that this electric double layer stores electrical energy through what is called double layer capacitance (Shi, 1996). This capacitance is not constant and can be found by the following equation:

$$C = \frac{\mathrm{d}\sigma}{\mathrm{d}\Psi} \tag{5}$$

Where σ is the charge of the surface and Ψ is the electrical potential at the surface of the activated carbon. As there is no simple method of measuring this surface charge and surface potential, it is better to view more general properties to which the double layer contributed, such as the battery charge capacity and the terminal voltage measured from the battery. The presence of activated-carbon would theoretically cause some capacitative effects in the battery due to the presence of the EDL. However, as in a capacitor, these process would only be expected to dominate the current and voltage at the beginning of a discharge, as eventually intercalation reactions (which run slower and are exhausted later) would become more prevalent. As such, the initial current and voltage during a discharge could be considerably higher than the values determined for most of the discharge, but the values would eventually stabilize when intercalation effects become more dominant.

A zinc-ion/graphite battery has a theoretical standard cell potential of 2.2 V (H. Zhang et al., 2020). Subtracting the oxidation potential of zinc at 0.76 V gives an standard reduction potential of 1.44 V for the intercalation of zinc in graphite. On the other hand, the reduction potential of manganese dioxide depends on the specific structure, and can vary between 1.3 and 1.4 V (Ming et al., 2019). Since the reduction potential of the zinc-graphite intercalation relation is similar to the reduction potential of

the zinc-manganese dioxide intercalation reaction, it can be assumed that the theoretical cell potential, even with the manganese dioxide, is still 2.2 V.

However, the values listed above for reduction potentials and the overall cell potential above do not consider variations from standard conditions, which require the Nernst equation:

$$E = E^\circ - \frac{RT}{nF} \ln Q \tag{6}$$

In this equation, E is the cell potential in non-standard conditions (measured in V), E° is the standard cell potential (V), R is the universal gas constant ($8.314\,\mathrm{J\,mol^{-1}\,K^{-1}}$), T is the absolute temperature, n is the number of electrons transferred in a single reaction, F is Faraday's constant, the absolute value of the charge of 1 mole of electrons ($96\,485\,\mathrm{C\,mol^{-1}}$), and Q is the reaction quotient for the specific reaction. For this specific type of battery, the net chemical equation can written as:

And, as known from (H. Zhang et al., 2020), the theoretical electrochemical potential from this net reaction is 2.2 V at the standard conditions. However, the concentration of the electrolyte can affect the cell potential through the Nernst eqution. Knowing that $n = 2$ for the zinc ion battery, the individual Nernst equations for the anode and cathode take the following form:

$$\begin{aligned} E_a &= {E^\circ}_a + \frac{RT}{2F} \ln \frac{[\mathrm{Zn^{2+}}]}{[\mathrm{Zn}]} \qquad \text{(Anode)} \\ E_c &= {E^\circ}_c + \frac{RT}{2F} \ln \frac{[\mathrm{ZnC}_x]}{[\mathrm{Zn^{2+}}]} \qquad \text{(Cathode)} \end{aligned} \tag{7}$$

However, the concentration of zinc and the concentration of the zinc-graphite intercalation compound is constant, as these materials are solids. Thus, they can be omitted from the calculation to give the following equations:

$$\begin{aligned} E_a &= {E^\circ}_a + \frac{RT}{2F} \ln[\mathrm{Zn^{2+}}] \\ E_c &= {E^\circ}_c - \frac{RT}{2F} \ln[\mathrm{Zn^{2+}}] \end{aligned} \tag{8}$$

The overall cell potential is equivalent to the difference in the reduction potentials of the cathode and anode. Thus, an overall Nernst equation can be derived for the cell, as shown in Equation 9.

$$
\begin{aligned}
E_{\text{cell}} &= E_{\text{cathode}} - E_{\text{anode}} \\
E_{\text{cell}} &= \left(E^{\circ}{}_{c} - \frac{RT}{2F}\ln[\text{Zn}^{2+}]\right) - \left(E^{\circ}{}_{a} + \frac{RT}{2F}\ln[\text{Zn}^{2+}]\right) \\
E_{\text{cell}} &= E^{\circ}{}_{c} - E^{\circ}{}_{a} - \frac{RT}{F}\ln[\text{Zn}^{2+}] \\
E_{\text{cell}} &= E^{\circ}{}_{\text{cell}} - \frac{RT}{F}\ln[\text{Zn}^{2+}]
\end{aligned}
\tag{9}
$$

However, since the zinc ions should be reacting at the cathode at around the same rate as the anode, the concentration of zinc ions is not changing much overall and the cell potential for a single battery should not be expected to change much. Furthermore, the quantity $\frac{RT}{F}$ is very small, so the deviation from the standard cell potential should be minimal. Thus, the Nernst equation's (Equation 6) effects can be ignored and the theoretical cell potential can be assumed to be 2.2 V for the batteries made in this experiment (H. Zhang et al., 2020).

The potential measured by placing a voltage probe across the ends of a battery while discharging would not be the cell potential however; every battery has an internal resistance, which is the opposition to the movement of charge within the battery. The cell potential is also called the emf, while the voltage while discharging is called the terminal voltage, and varies with the current through the following equation:

$$
V = \varepsilon - Ir \tag{10}
$$

In this equation, V is the terminal voltage, ε is the emf (or cell potential), I is the current, and r is the internal resistance of the battery. Though modeled as a resistor, the internal resistance is not caused by any resistive elements in the battery; rather, it is caused by the formation of concentration gradient that opposes the flow of current (T. Zhang et al., 2020).

A nickel-metal hydride (NiMH) battery was used as standards for comparison to the zinc-ion battery, as its uses overlap with many of the potential uses of a zinc-ion battery. NiMH batteries are a very common type of rechargeable battery. They are able to safely discharge at higher currents, can be charged and discharged hundreds of times with minimal loss in capacity, and remain usable for a long amount of time. However, NiMH batteries also have the highest rate of energy loss when not in use of the common batteries. This phenomenon is known as self-discharge, and standard NiMH batteries can lose

15% of their stored energy per month, making these batteries ineffective for grid storage. Nevertheless, NiMH are useful for The reactions of the nickel-metal hydride battery can be seen below. In these equations, "M" represents a metal alloy of some sort. NiMH cells require an alkaline electrolyte, which usually is potassium hydroxide.

$$\begin{aligned} H_2O + M + e^- &\rightleftharpoons OH^- + MH \qquad \text{(Anode)} \\ Ni(OH)_2 + OH^- &\rightleftharpoons NiO(OH) + H_2O + e^- \qquad \text{(Cathode)} \end{aligned} \tag{11}$$

In order to determine the feasibility of zinc-ion batteries for usage in commericial applications, the cathode and electrolyte were varied to find which combinations would produce the best results. In every design, the zinc anode was kept the same, and the casing used was identical. To provide a basis for comparison, a nickel-metal hydride battery was discharged as well.

Materials and Methods

Materials

To design a zinc-ion battery, a cathode, anode, electrolyte, casing, and separator each must be created and combined into a final structure. The experimental casing was a 3D printed hollow cylinder with a 5.08 cm diameter and a height of 10.16 cm, as shown in Figure 3. While larger than a standard battery, this size allowed for easier designing and testing. The casing was made from PETG (polyethylene terephthalate glycol-modified), though any plastic used in 3D printing would have been suitable.

Figure 3: The casing used to house the battery.

The cathode, or positive terminal, was constructed from graphite current collector and a carbon-electrolyte mix, with manganese dioxide added for the final version. Three different current collectors were used for different parts of the experiment. A total of 66.23 g of activated carbon, 36.22 g of graphite

powder, and 6.47 g of manganese dioxide were used to build the cathodes. The specific activated carbon powder used, as shown in Figure 4, was coconut-based activated carbon powder from Belle Chemical, obtained from Amazon. The graphite powder, shown in Figure 5, was obtained from Flinn Scientific through Dallastown Area High School. The manganese dioxide is shown in Figure 6 and was obtained through Amazon as well.

Figure 4: The activated carbon.

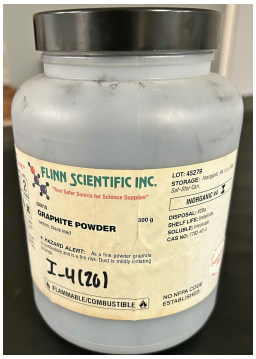


Figure 5: The graphite powder.

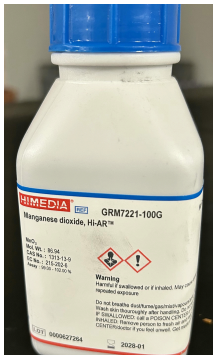


Figure 6: The manganese dioxide.

The electrolyte was also varied between designs to determine whether the anion plays a significant role in the battery. For the designs described in this paper, zinc sulfate heptahydrate and zinc chloride were used. Zinc sulfate is more stable and is often used in zinc-ion battery prototypes, but zinc chloride has a much higher solubility, allowing for higher concentrations and thus higher currents. All of the zinc sulfate and zinc chloride used in these experiments was obtained via Amazon, with the specific brands visible in Figure 7 and Figure 8, respectively. A total of 82.85 g of zinc sulfate heptahydrate and 136.29 g of zinc chloride were used to produce the various electrolytes used for the batteries described in this paper.

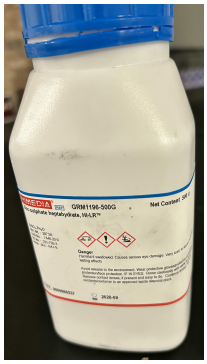


Figure 7: The zinc sulfate heptahydrate used in this paper.

Figure 8: The zinc chloride used in this paper.

The anode used in these batteries was a zinc foil, and was kept constant through every iteration. Since the purpose of the anode of a zinc-battery is to simply oxidize zinc, any other material would be less effective. The zinc foil is shown in Figure 9, and a new piece of foil was used in each iteration. The foil had a thickness of 0.1 mm, and a 10 mm by 8 mm piece was cut out for each iteration.

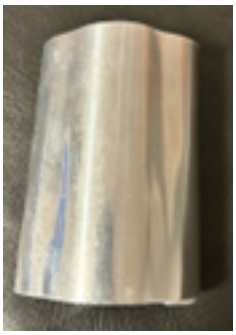


Figure 9: A piece of zinc foil used in the designs.

To carry out the measurements necessary to evaluate the different batteries, a Vernier current and voltage probe, as shown in Figure 10 and Figure 11, were wired into the various circuits. Since the data often needed to be collected unattended, a LabQuest 2, as shown in Figure 12, was used to collect the data.

Figure 10: A current probe.

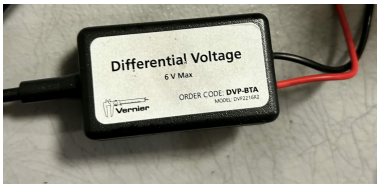


Figure 11: A voltage probe.

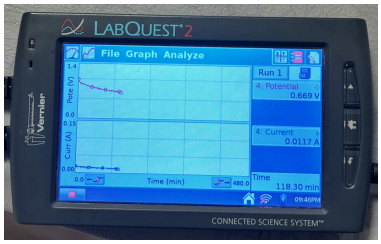
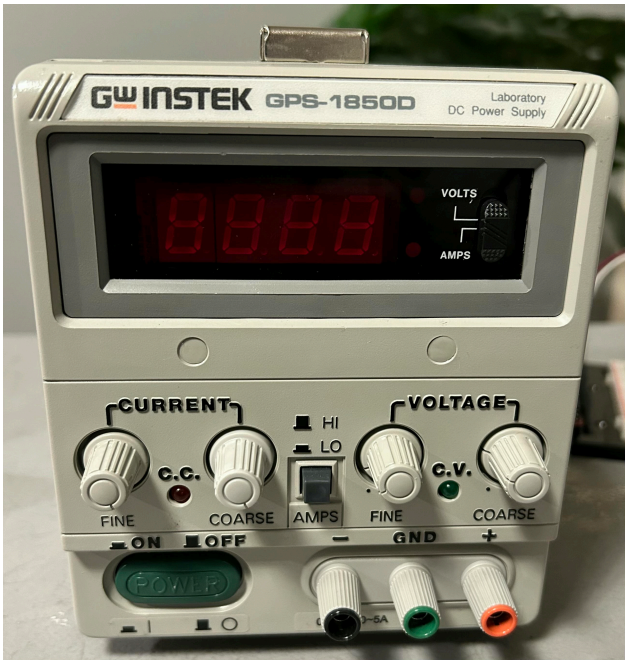


Figure 12: A Vernier LabQuest 2, used to record data unattended.

In order to charge the various batteries, a power supply was used with constant voltage control. The specific power supply used, as shown in Figure 13, was a GW Instek GPS-1850D DC power supply.



Procedure

While the quantities and some materials differed between the various iterations, the fundamental methodology remained mostly unchanged. Thus, this paper has described the first iteration in detail, while only listing the most notable changes between iterations, as well as any differences in the testing done for each iteration.

The design of each battery was divided into three steps: making the electrolyte, constructing the cathode, and building the battery. The electrolyte used in the first battery consisted of 2.5 M $ZnSO_4$ solution. 40.37 g of $ZnSO_4$ were measured out and placed in a beaker. Regular water was added until the volume of the solution in the beaker was 100 mL to produce the 2.5 M solution. The solution was thoroughly stirred until clear. Afterward, 15.56 g of activated carbon powder and 6.33 g of graphite powder were measured out and stirred in a bowl until thoroughly mixed. Next, a pipette was used to slowly add small amounts of the zinc sulfate solution to the powder mix. The mixture was made damp, yet not wet, and had a consistency similar to peanut butter when complete. This paste is shown in Figure 14.

Figure 14: A picture of the carbon-electrolyte paste.

The paste was then spread evenly around a rod of graphite to produce the cathode. However, to stop the paste from interacting with the anode and dissolving into the electrolyte, the cathode was wrapped with filter paper and sealed with hot glue. The cathode was placed in the battery housing (Figure 3) with the zinc foil. Lastly, the remaining amount of solution was added to the casing to act as the electrolyte. The finished version of the first battery is shown in Figure 15.

Figure 15: The completed first battery. The cathode can be seen sticking out of the battery, while the zinc foil anode lines the edge of the casing.

The first iteration of the battery was a prototype and not used for any of the data shown below, as the data was neither standardized nor reliable. The second iteration of the battery was similar to the first, but involved proper standardization of data collection, and the data from the second iteration was used for analysis.

The second iteration of the battery, though similar to the first battery, was using less materials, so that a general decrease was expected across the measurements. The electrolyte was made using 21.6 g

of $ZnSO_4$, and water was added until the solution had a volume of 50 mL, equivalent to a concentration of 1.5 M. Only 9.22 g of activated carbon and 3.74 g of graphite powder were used. The graphite rod from the first iteration was cleaned off and reused for the second iteration.

The battery was connected in series with a current probe and a resistance box (which allows for varying the resistance), while a voltage probe was connected in series. The circuit is shown in Figure 16, though a schematic is also visible in Figure 17. The Vernier LabQuest 2 (shown in Figure 12) was set to measure 30 samples per minute for 480 minutes, which is equivalent to 1 sample every 2 seconds for 8 hours.

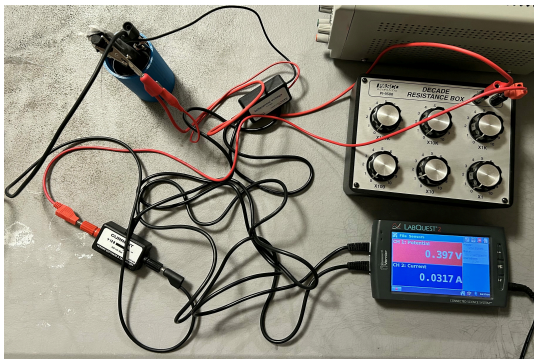


Figure 16: An example of a discharge circuit.

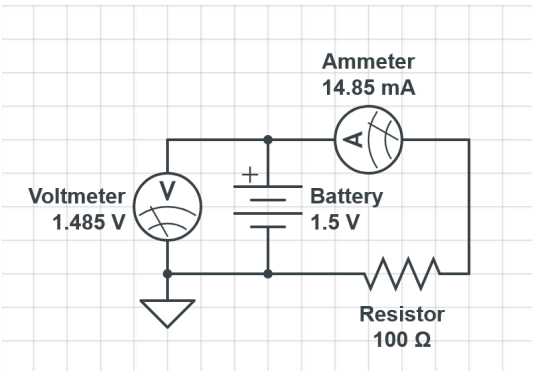


Figure 17: A schematic of a discharge circuit. The battery shown has internal resistance.

The next step was to charge the battery. Simply charging the battery while measuring the current and voltage would not show the emf, which is equivalent to the open circuit voltage. To do so, an Arduino microcontroller board was connected to a relay wired into the circuit. The arduino and its wiring when not connected to a battery circuit is shown in Figure 18.

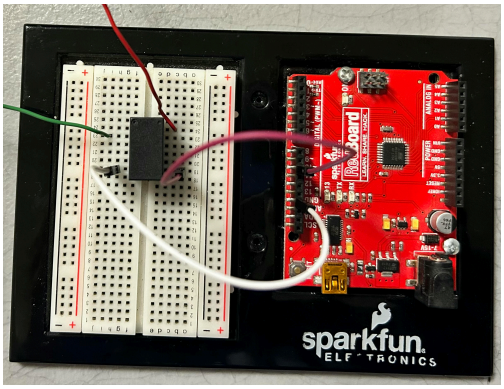


Figure 18: The arduino used to time the charging circuit. The purple wire connects digital pin 10 to the coil and returns to ground through the white wire. The red and green wires connect to the battery circuit.

The arduino was set to wait for 4 minutes, disconnect the circuit for one minute, then reconnect the circuit, on loop. The code is shown here:

```
int relayPin = 10;

void setup() {
  Serial.begin(115200);
  pinMode(relayPin, OUTPUT);
}

void loop() {
  digitalWrite(relayPin, LOW);
  delay(240000);
  digitalWrite(relayPin, HIGH);
  delay(60000);
}
```

This program sets the output to digital pen. The `setup()` function runs once when the Arduino is first connected to power. It initializes the Arduino to read any input data at 115200 symbols per second (115200 baud) and sets digital pin 10 as an output. Afterword, in the `loop()` function, the Arduino turns off the pin, waits 4 minutes, turns the pin on, waits for a minute, and then repeats. Turning the pin on activated the relay and opens the circuit. The LabQuest 2 was timed to collect the current and voltage once very 2.5 minutes, so that the current while charging could be read, then the emf could be read while

the circuit was open, 2.5 minutes later. The LabQuest 2 was set to collect, then the Arduino was activated 10 seconds later, so that the two would not interfere with one another. Each charge lasted for 60 minutes, with the current supply set to a constant voltage of 1.5 V. This time and voltage was chosen to ensure that the battery would not be charged for too long (leading to battery damage and the formation hydrogen gas). A schematic of the charging circuit is shown in Figure 19, where the battery is modeled as a load (resistance) for the power supply.

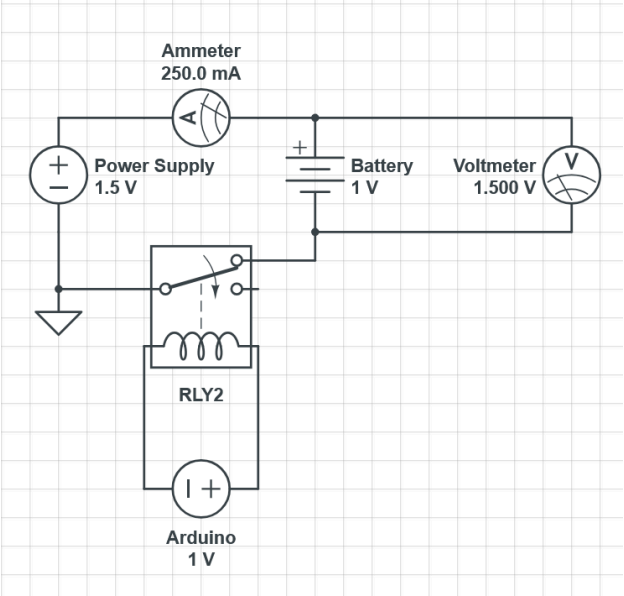


Figure 19: A schematic of the battery while charging.

The second iteration battery was discharged across a 100 Ω resistor three times and charged three times.

The third iteration of the battery featured a different graphite rod and a switch to zinc chloride electrolyte. Anhydrous zinc chloride was used, as shown previously in Figure 8. Equivalent amounts of activated carbon and graphite powder were used as was in the second iteration of the battery. However, the electrolyte was instead made using 20.45 g of $ZnCl_2$ to produce 1.5 M $ZnCl_2$ solution. This solution was used in the creation of the paste and as the electrolyte for the battery.

The circuits used to test the third battery were the same as those used to test the second iteration battery. However, the external resistance was varied for each discharge. The third battery

battery was originally discharged across 10 Ω, then at 1 000 Ω, and lastly at 100 Ω. This battery was charged between each discharge using the process listed before at 1.5 V.

As an extra test, the cathode from the third iteration battery was combined with a new piece of zinc in zinc sulfate solution instead to examine the effect of mixing the electrolytes. Water was added to 40.91 g of zinc sulfate until the total concentration of the solution reached 100 mL. This electrolyte was then used in place of the zinc chloride previously used for the third iteration. This battery was only discharged once, however, due to the terrible performance of this battery.

The fourth iteration of the battery was designed with much more graphite. A new cathode consisting of a fresh graphite rod, 9.88 g of activated carbon and 11.19 g of graphite powder were used, while 40.89 g of zinc chloride were used instead to increase the concentration to 3 M $ZnCl_2$. This battery was discharged at 10 Ω, 1000 Ω, and 100 Ω, with the battery being charged between each discharge.

The fifth and final iteration of the battery involved some more drastic changes, in a last attempt to improve performance and show more "battery-like" behavior. First, distilled water was obtained and used instead of regular water. The concentration was increased up to 6 M, using 102.23 g of $ZnCl_2$ combined with enough distilled water to create 125 mL of solution at 6 M concentration. The cathode was made using 16 g of activated carbon, 7.77 g of graphite powder, and 6.47 g of manganese dioxide through the same processes as before. By adding the manganese dioxide, it was thought that the battery may perform much better. Then, through the same processes as done before, the battery was constructed in the casing. To ensure that the final battery would show the best performance upon data collection, this battery was first charged for an hour at 2.4 V (the theoretical potential for this battery), where data was not collected. Data was then collected of the final battery being discharged across a 1000 Ω resistor for 240 minutes (4 hours) to allow the battery to "break-in" (allowing for an increase in performance after the first discharge, shown in some studies (Ming et al., 2019)) and provide optimal performance on the next run. Using the Arduino-based circuit from before, this battery was recharged for 1 hour at 2.2 V. Lastly, the battery was discharged across a 10 Ω resistor for 8 hours. While a lower resistance would have allowed for more energy to be released over the same period, keeping the resistance at 10 Ω allowed for fair comparison with all of the previous tests.

Finally, a standard rechargeable AA nickel-metal hydride battery was discharged to provide a comparison. The battery was discharged across a 1000 Ω resistor for 8 hours. Due to the dangers of recharging a commercial battery across a power supply and the inability to control the workings of a commercial NiMH battery charger safely, the battery was not recharged, and this graph was only used as a standard for comparison.

Results

Data

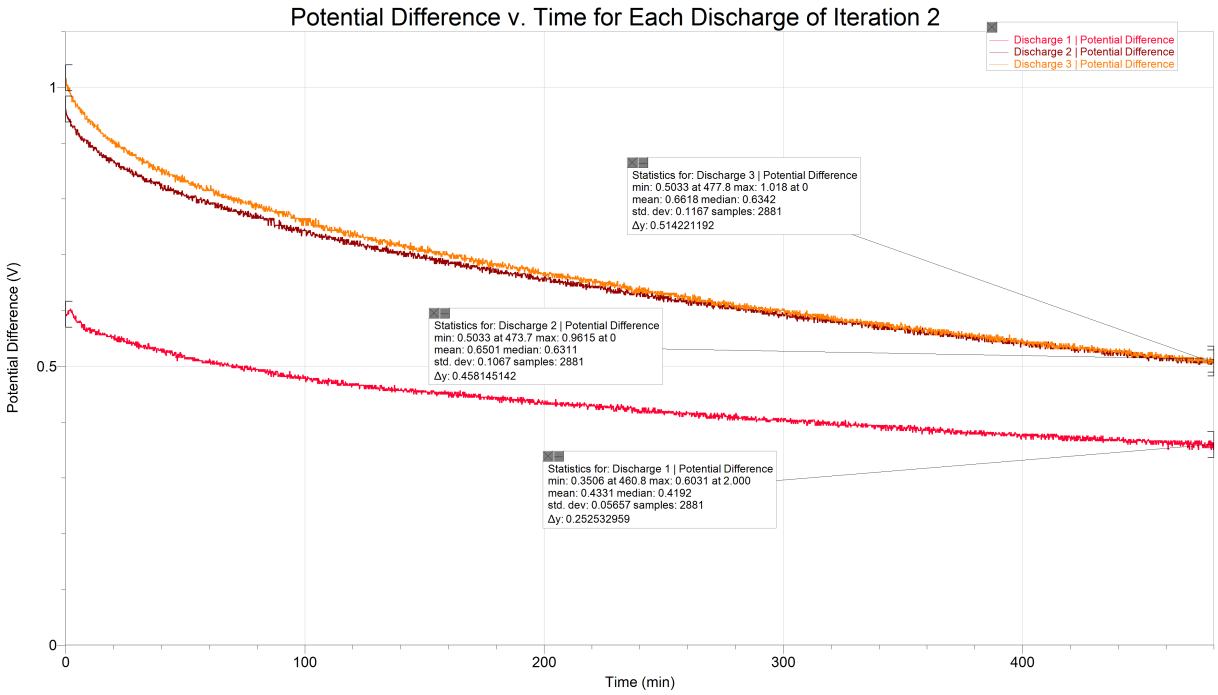


Figure 20: A graph showing how the voltage changed over time for each discharge of Iteration 2.

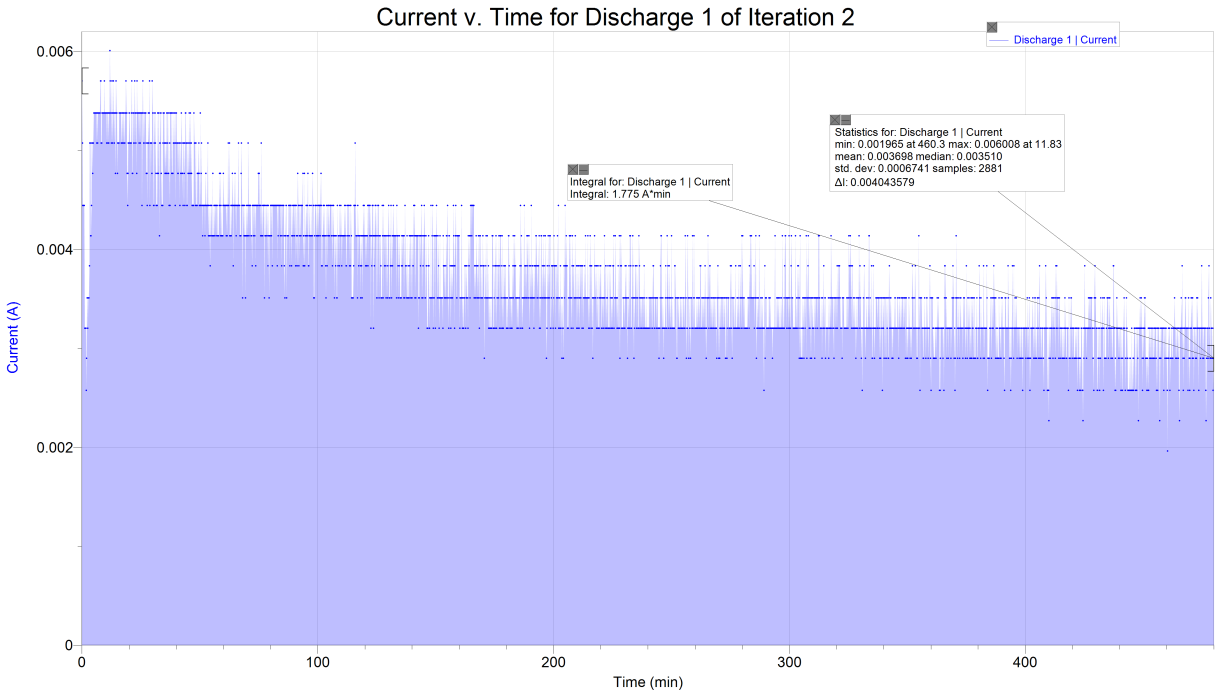


Figure 21: A graph showing current versus time and the integral of current for the first discharge of Iteration 2.

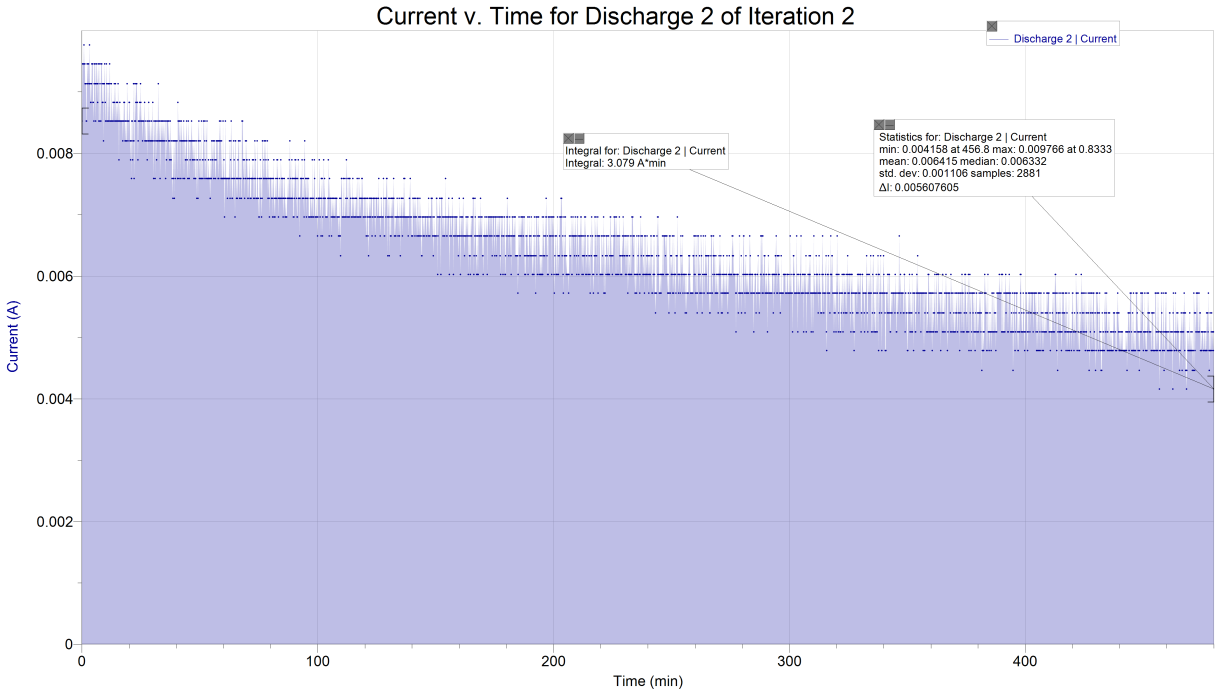


Figure 22: A graph showing current versus time and the integral of current for the second discharge of Iteration 2.

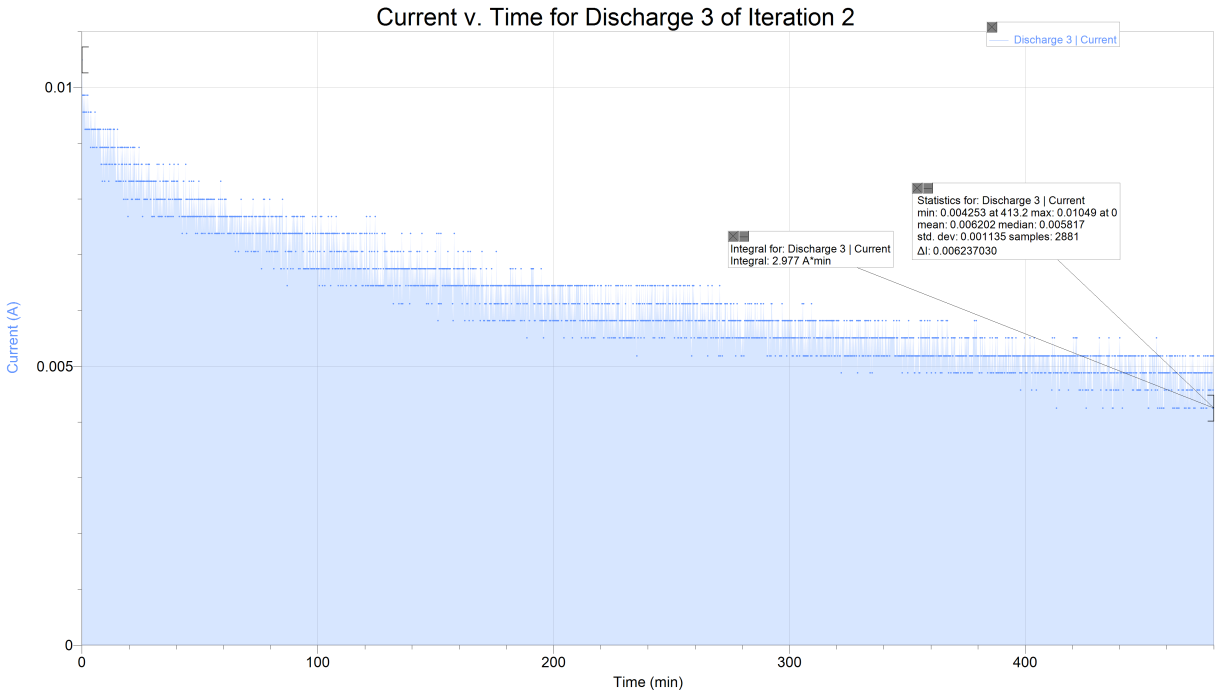


Figure 23: A graph showing current versus time and the integral of current for the third discharge of Iteration 2.

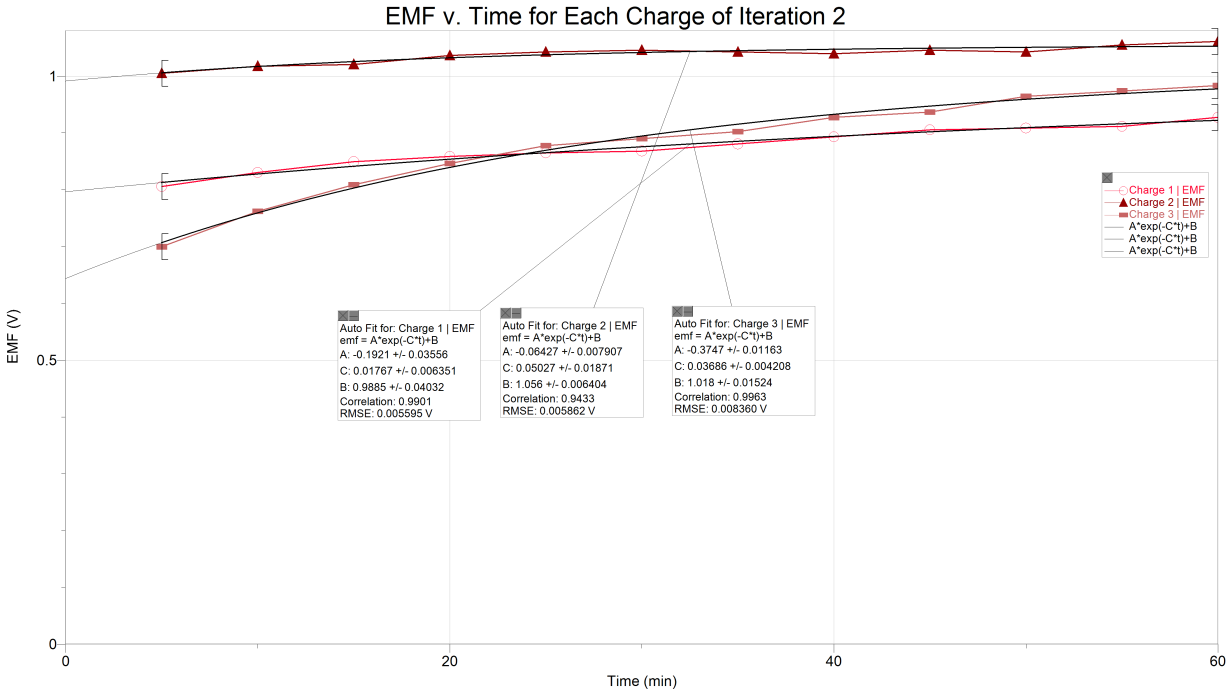


Figure 24: A graph of the emfs measured during each charge of Iteration 2.

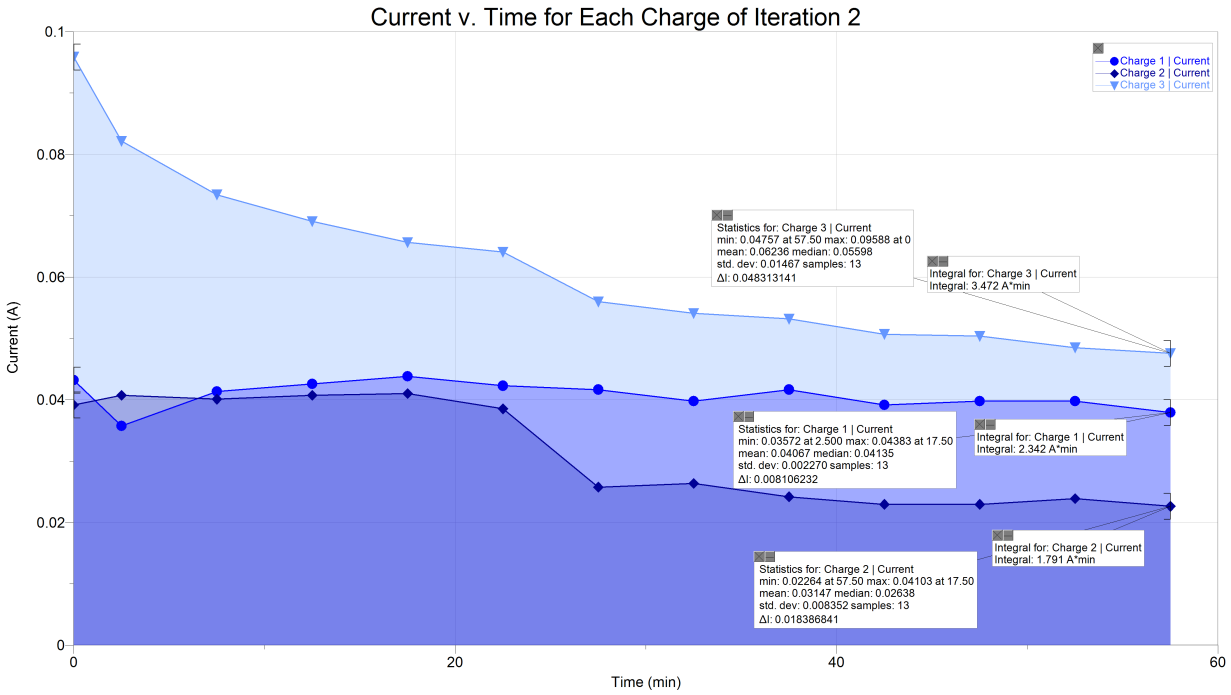


Figure 25: A graph of the currents measured during each charge of Iteration 2.

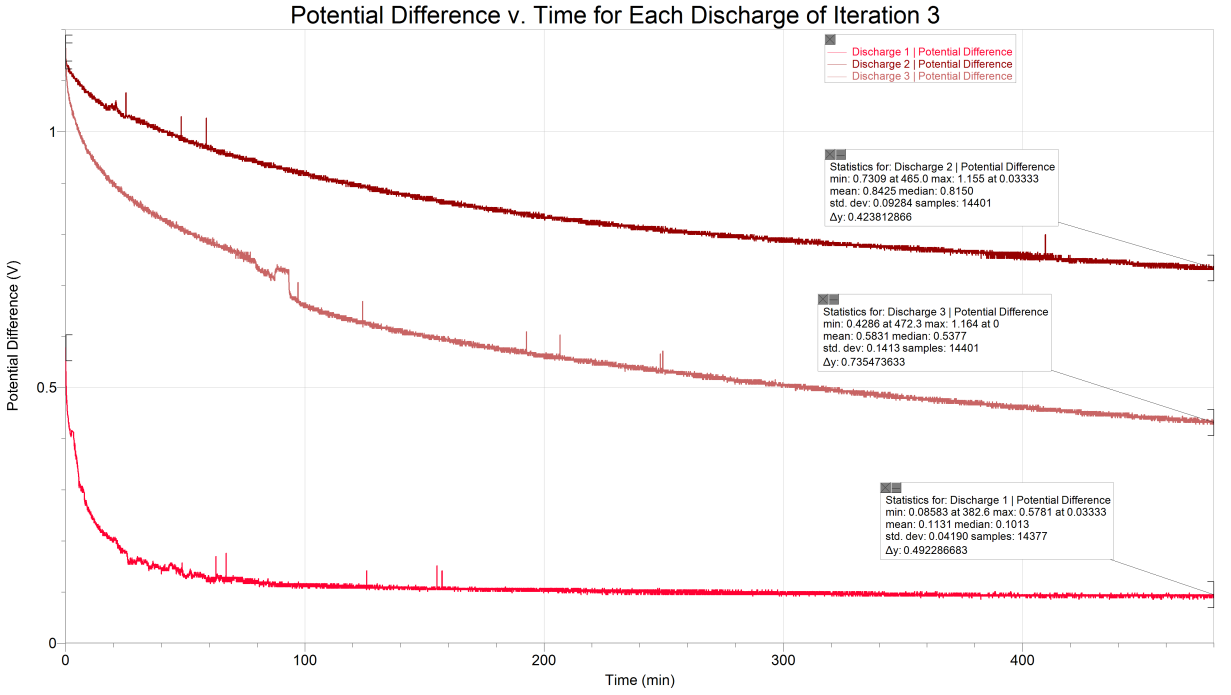


Figure 26: A graph showing how the voltage changed over time for each discharge of Iteration 3.

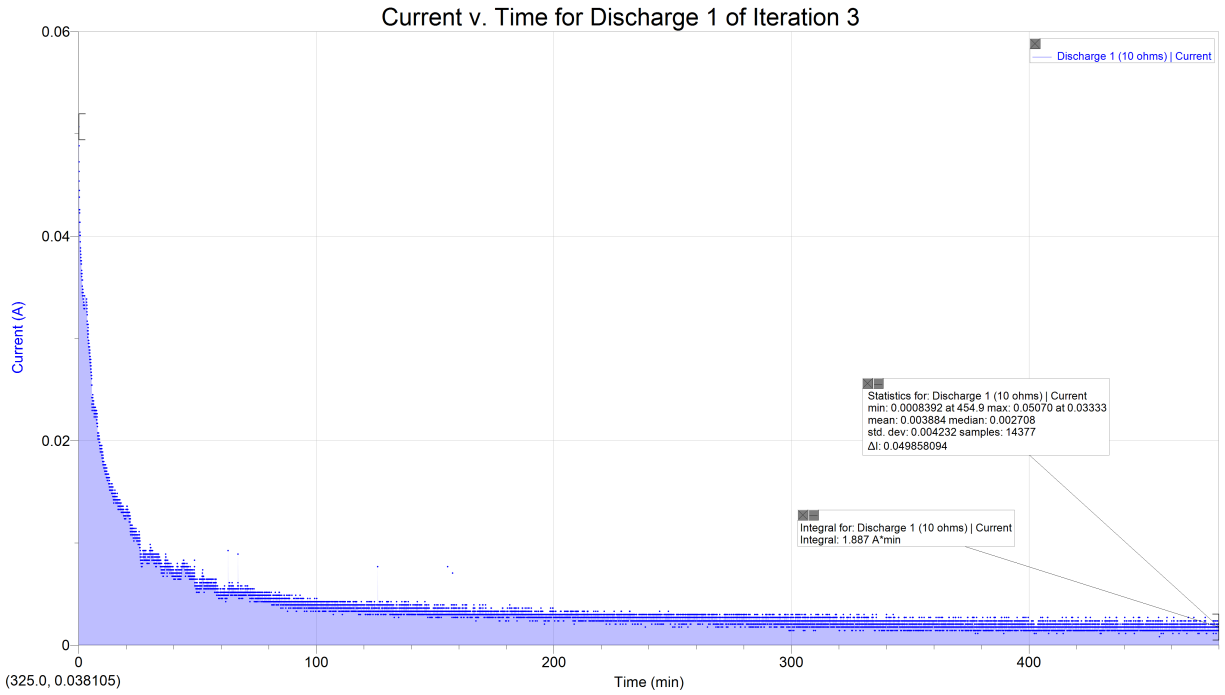


Figure 27: A graph showing current versus time and the integral of current for the first discharge of Iteration 3.

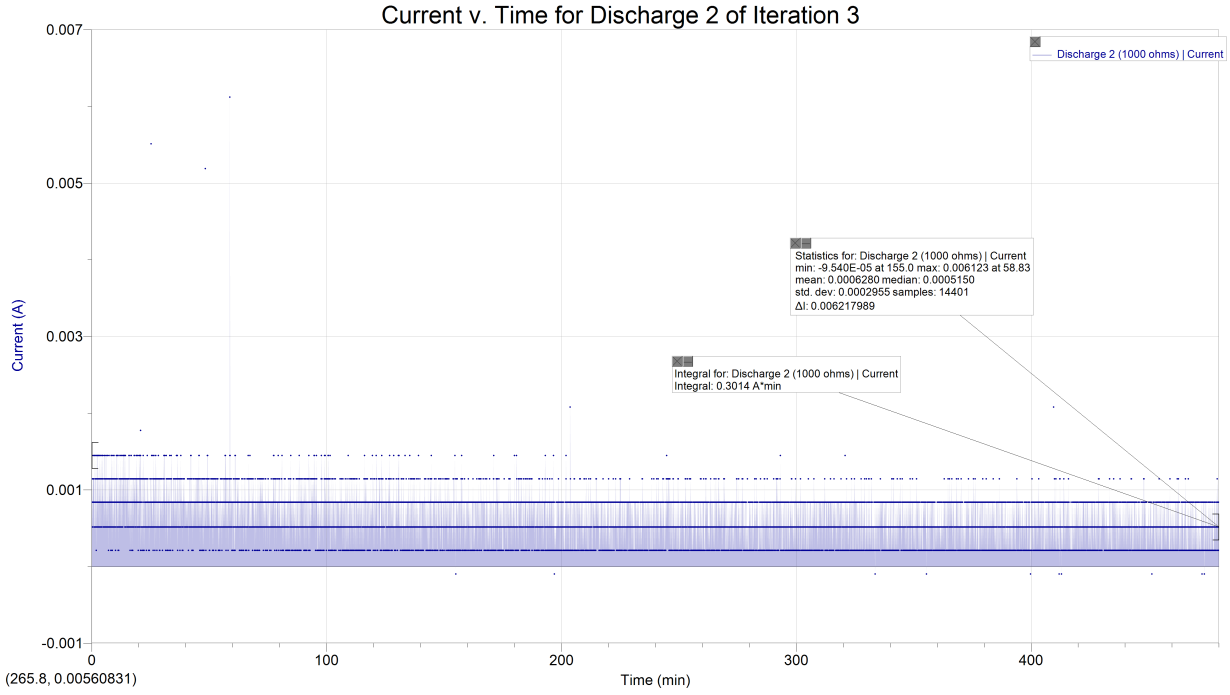


Figure 28: A graph showing current versus time and the integral of current for the second discharge of Iteration 3.

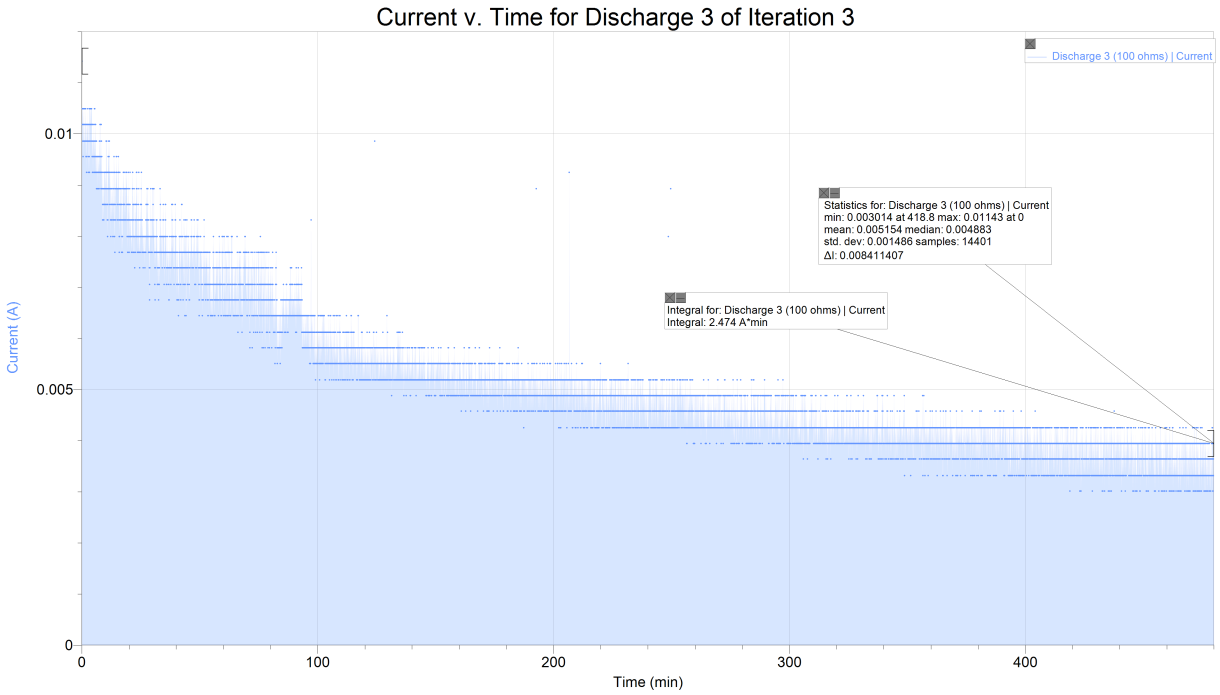


Figure 29: A graph showing current versus time and the integral of current for the third discharge of Iteration 3.

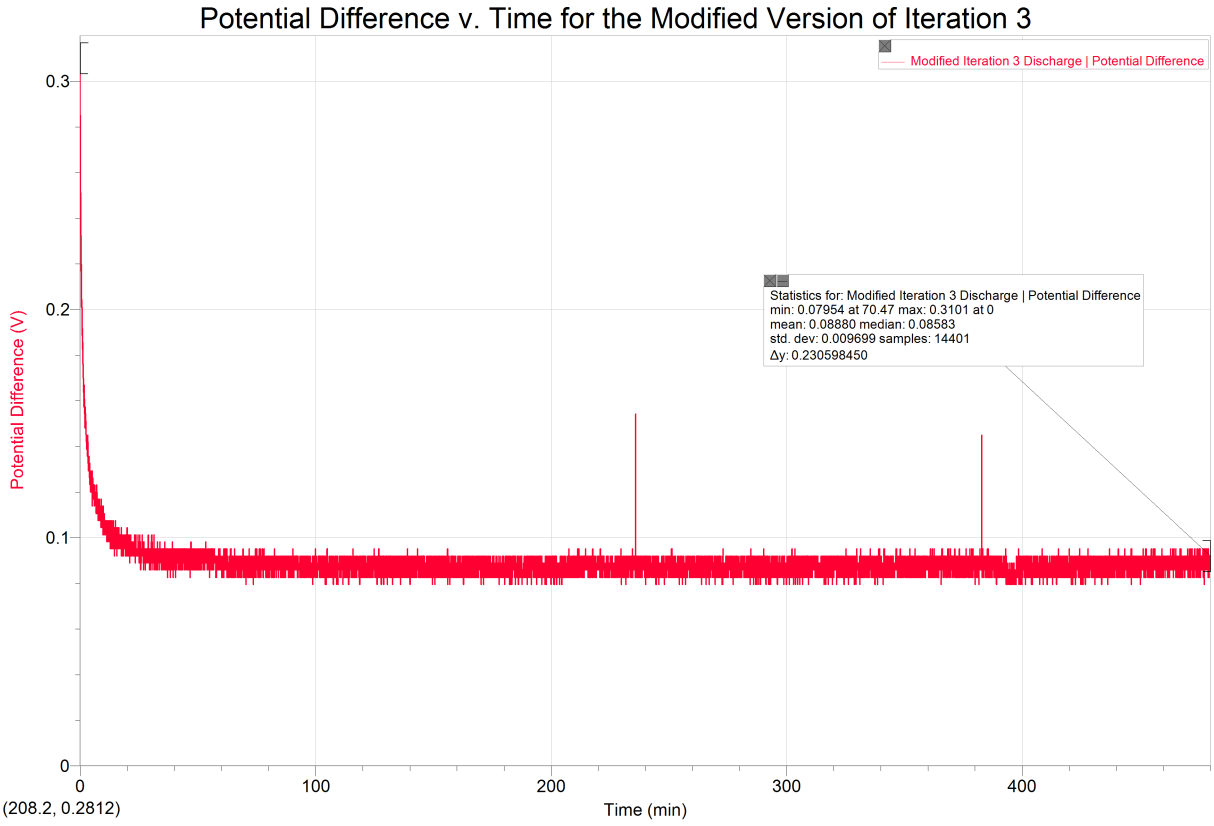


Figure 30: A graph showing how the voltage changed over time for the discharge of the modified Iteration 3.

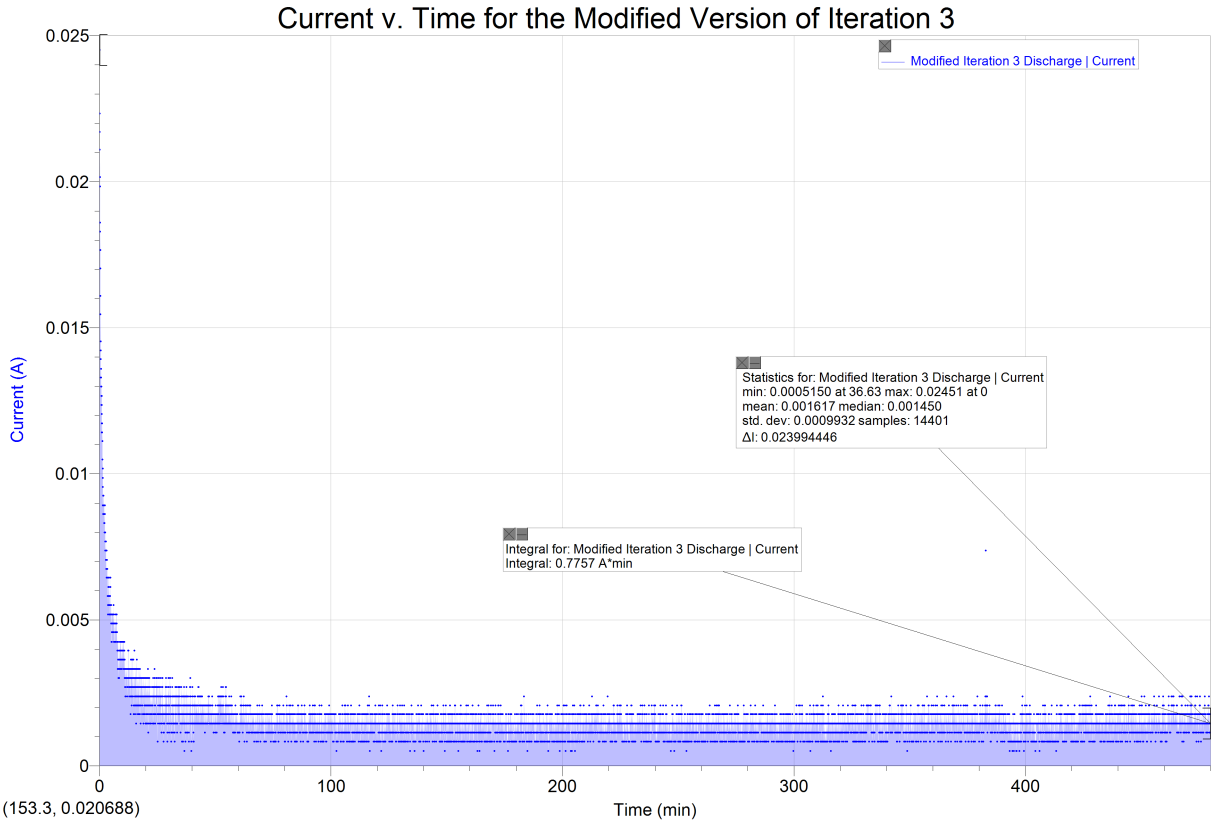


Figure 31: A graph showing how current varied over time and the integral of current for the modified Iteration 3.

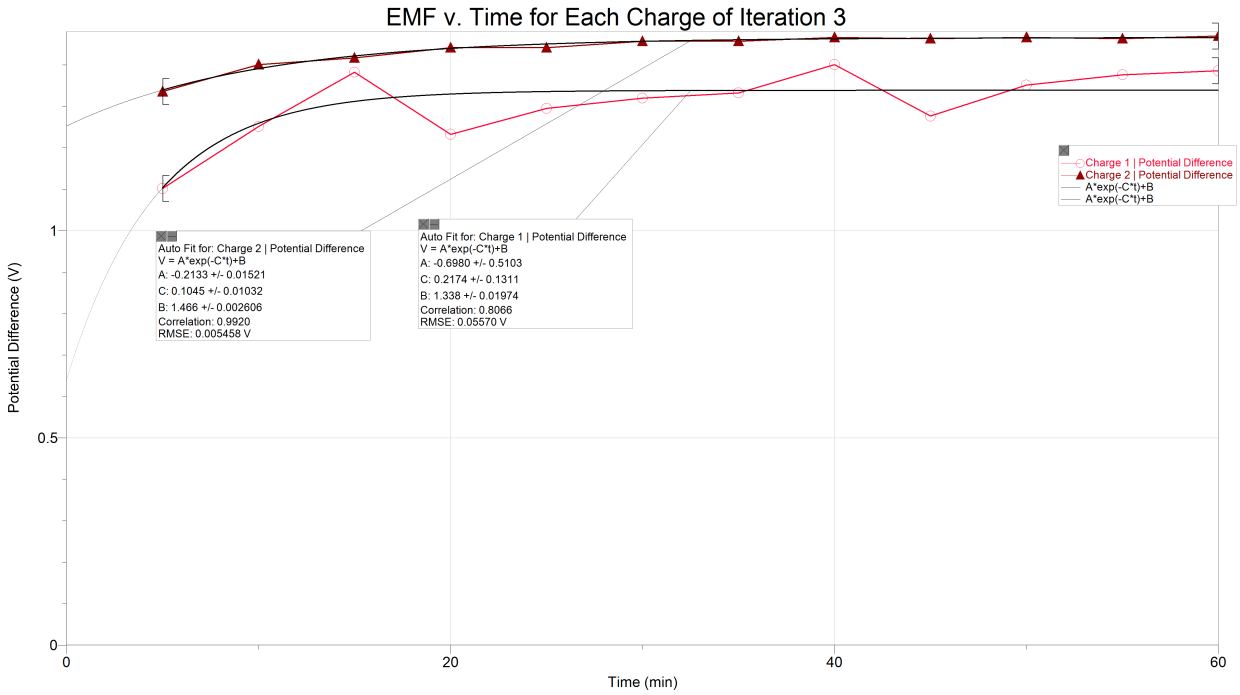


Figure 32: A graph of the emfs measured during each charge of Iteration 3.

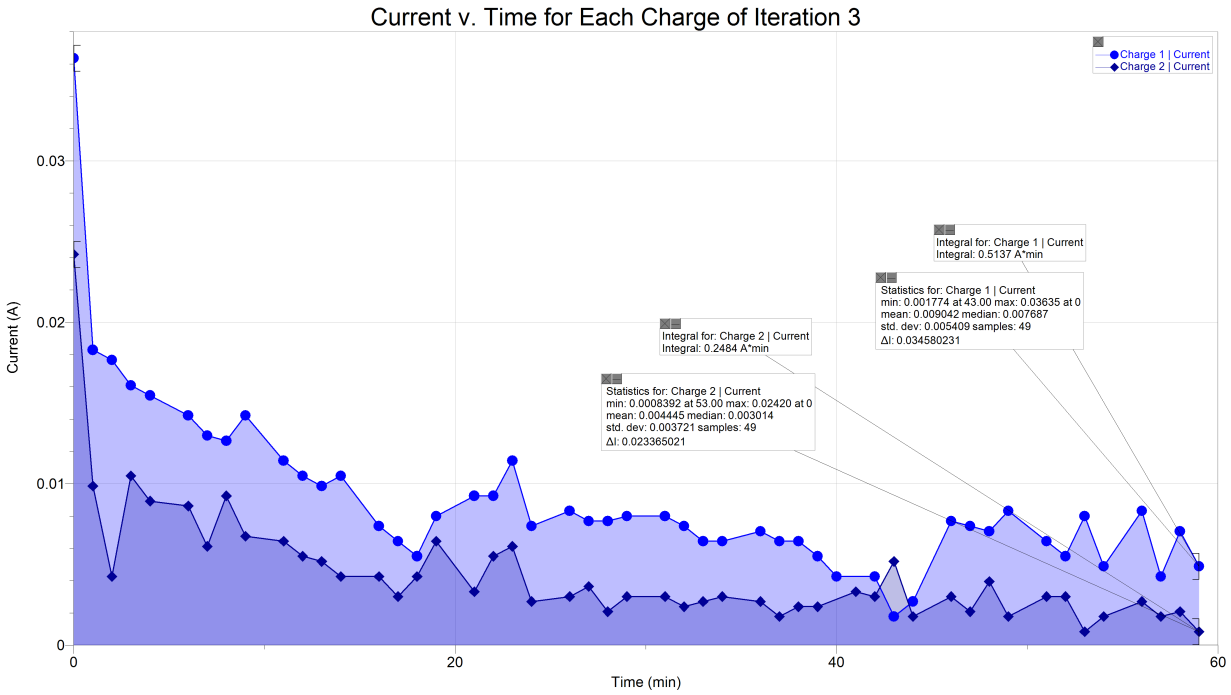


Figure 33: A graph of the currents measured during each charge of Iteration 2.

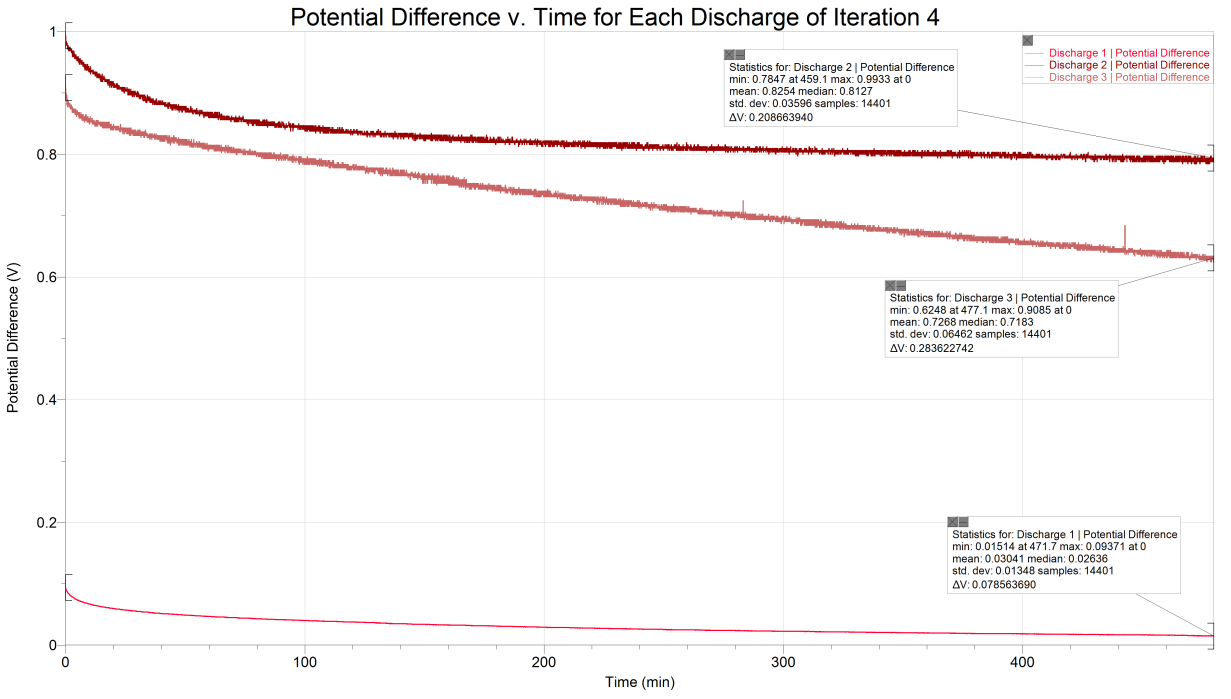


Figure 34: A graph showing how the voltage changed over time for each discharge of Iteration 4.

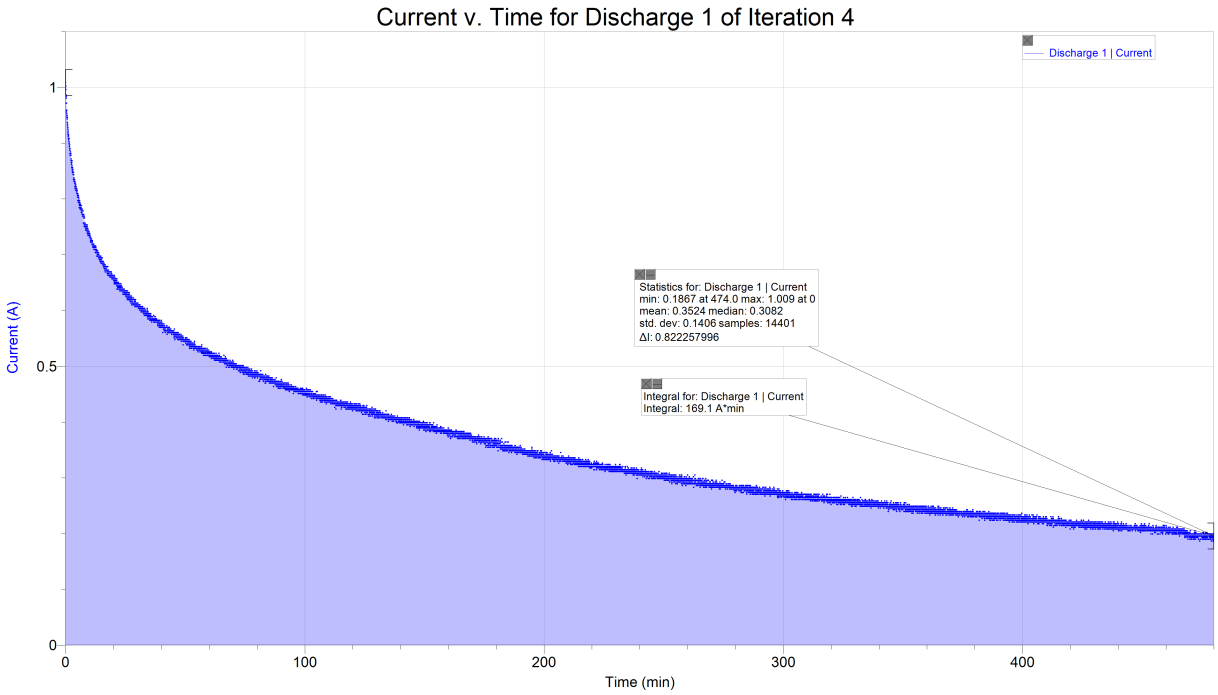


Figure 35: A graph showing current versus time and the integral of current for the first discharge of Iteration 4.

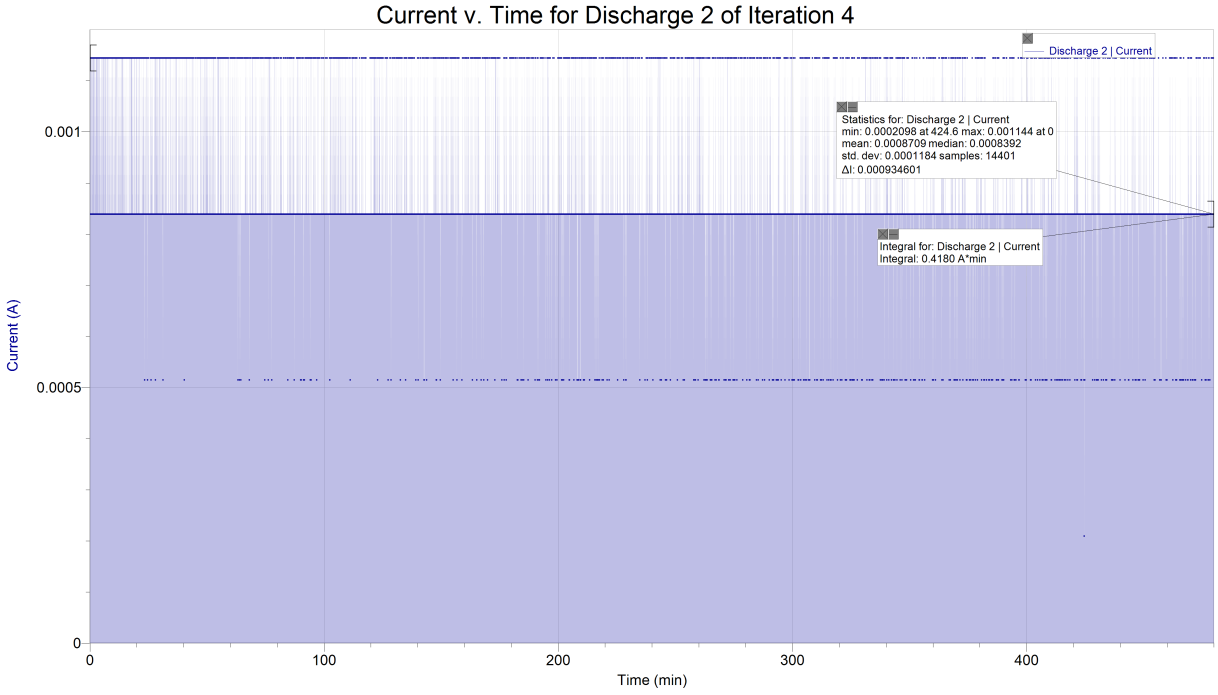


Figure 36: A graph showing current versus time and the integral of current for the second discharge of Iteration 4.

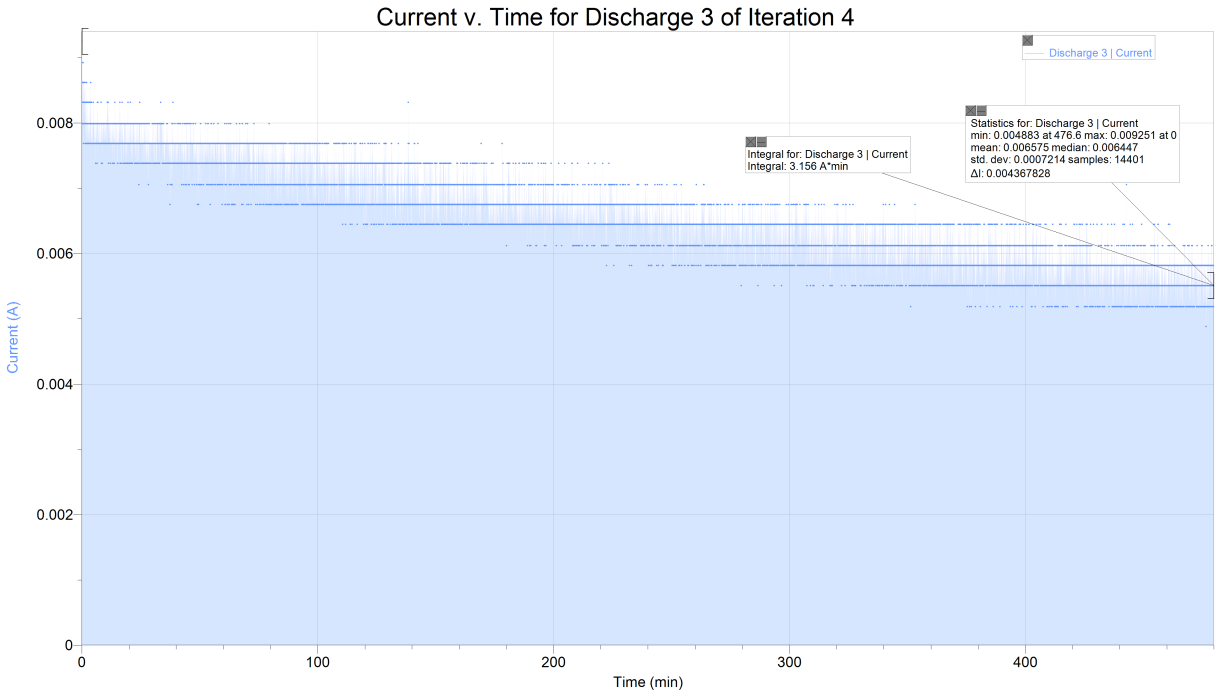


Figure 37: A graph showing current versus time and the integral of current for the third discharge of Iteration 4.

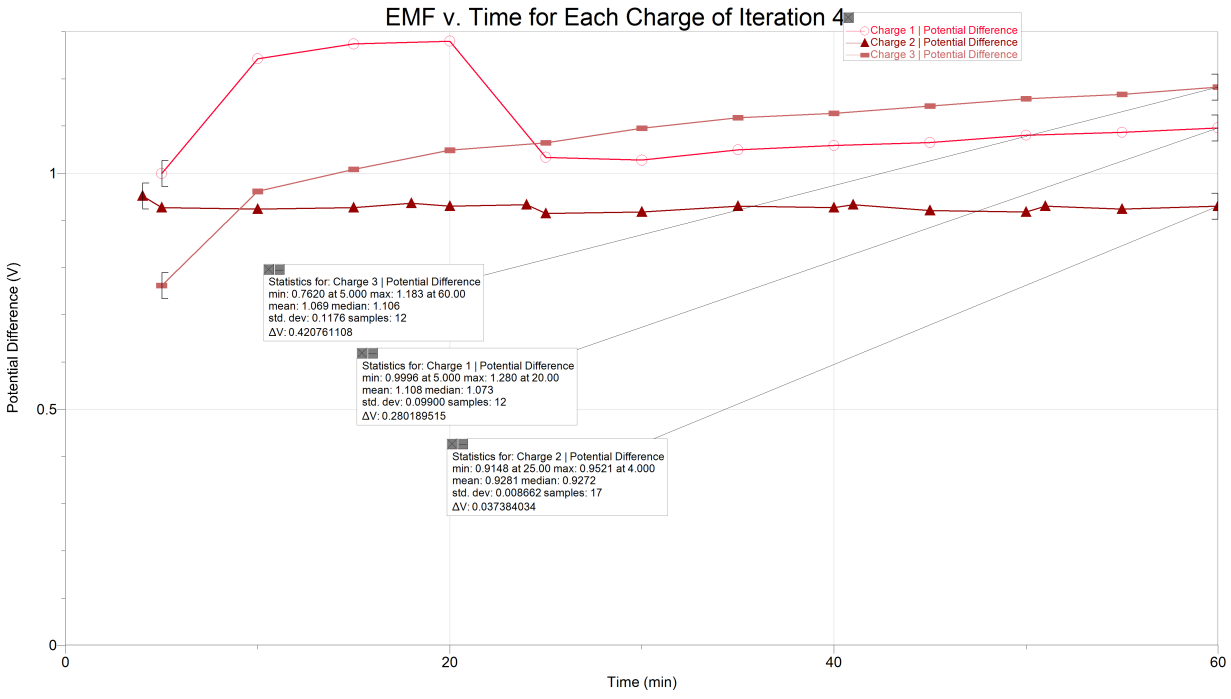


Figure 38: A graph of the emfs measured during each charge of Iteration 4.

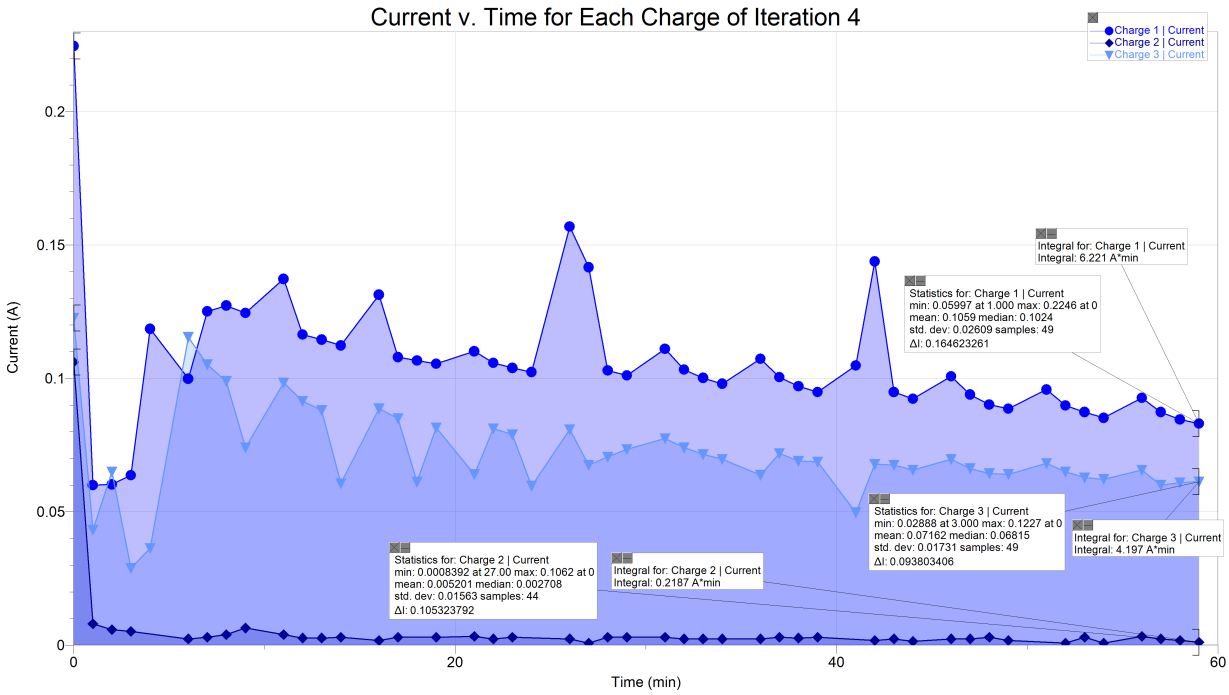


Figure 39: A graph of the currents measured during each charge of Iteration 2.

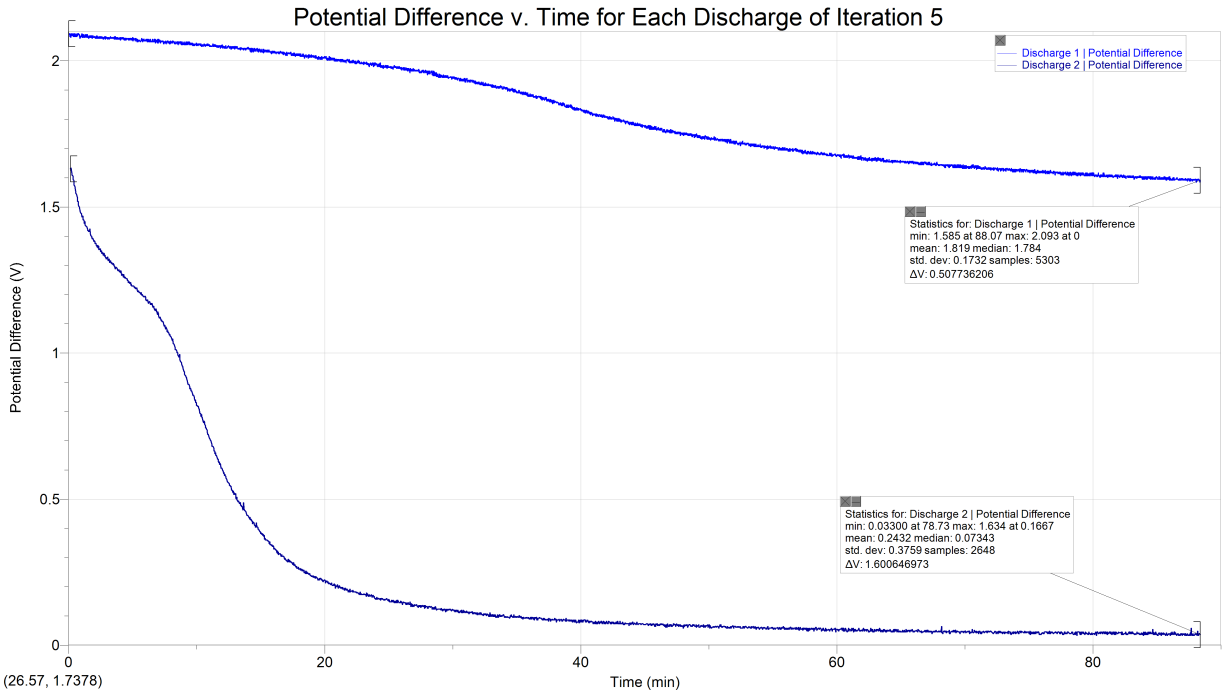


Figure 40: A graph showing how the voltage changed over time for each discharge of Iteration 5.

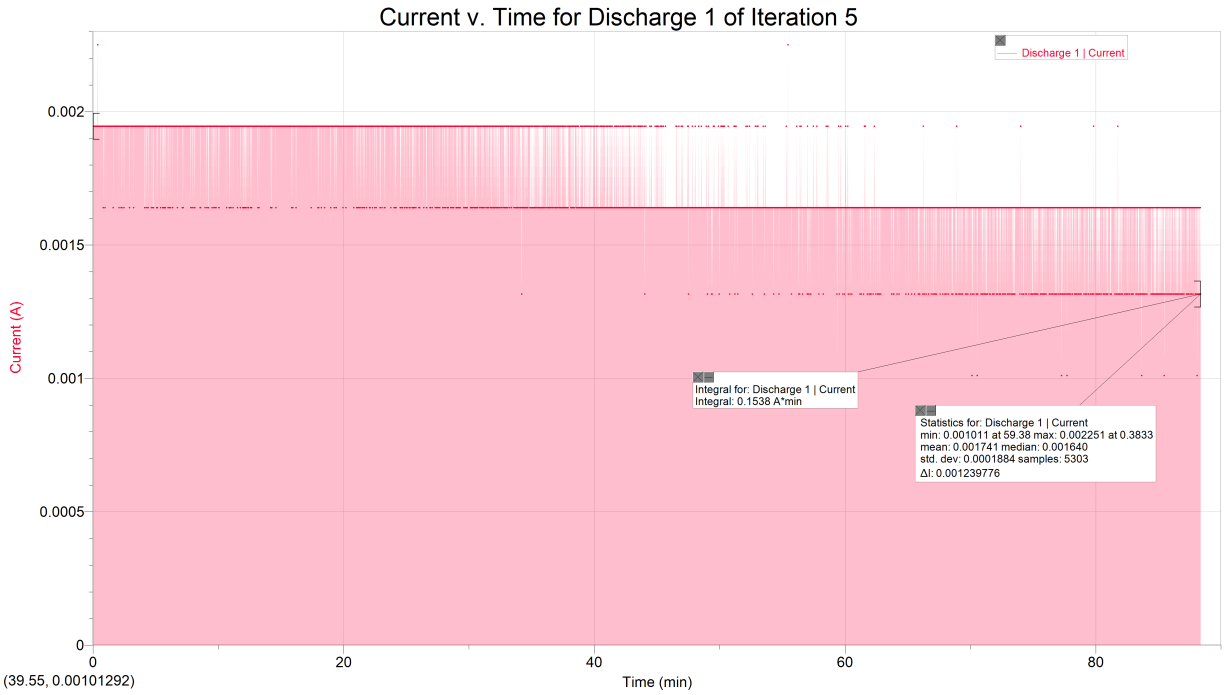


Figure 41: A graph showing current versus time and the integral of current for the first discharge of Iteration 5.

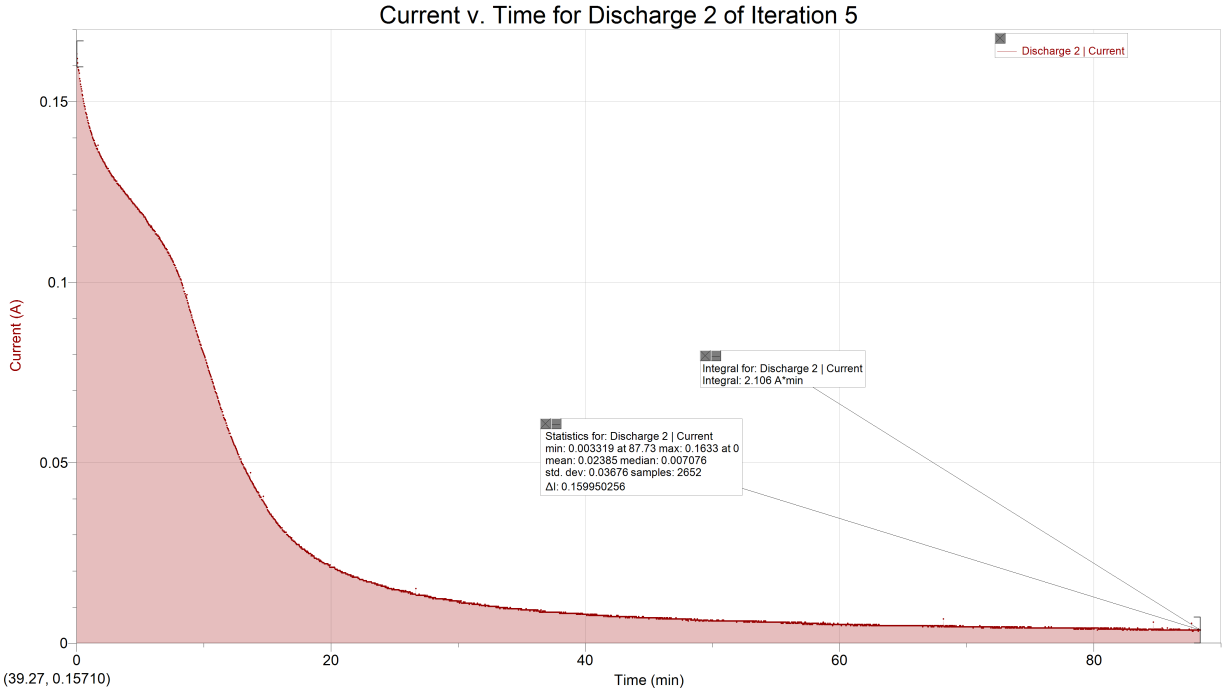


Figure 42: A graph showing current versus time and the integral of current for the second discharge of Iteration 5.

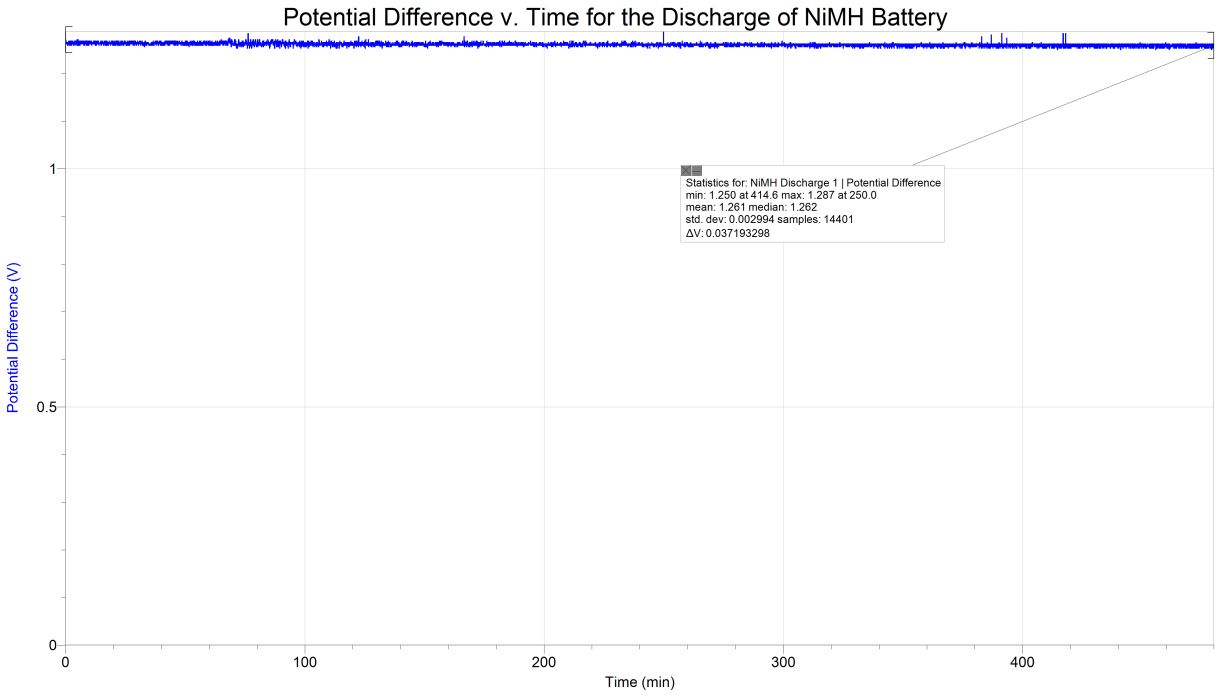


Figure 43: A graph showing how the voltage changed over time for the discharge of the NiMH battery.

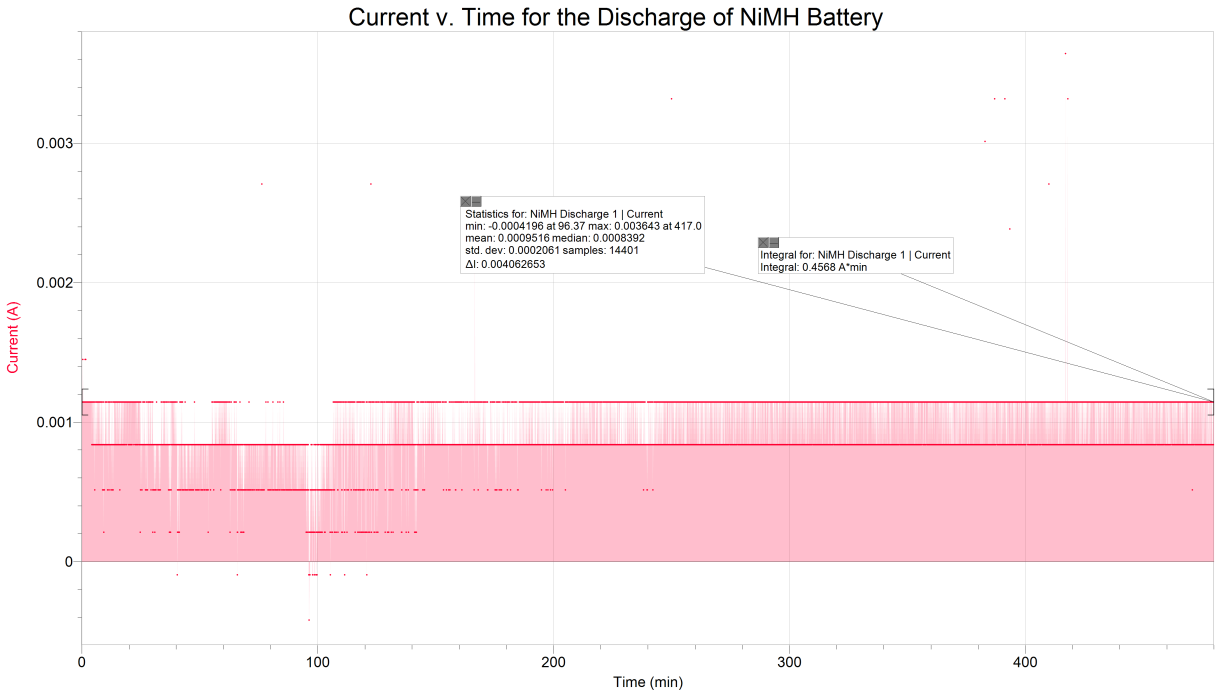


Figure 44: A graph showing current versus time and the integral of current for the discharge of the NiMH battery.

Data Analysis

The power from the battery at any given instant can be calculated using the voltage and the current using the following equation:

$$P = IV \tag{12}$$

Using Equation 12, the graph of power v. time can be plotted for all of the charges and discharges, as shown in the following figures. Using these graphs of power v. time, the total energy released by the battery over that time period can be calculated using the integral of power in the following equation:

$$E = \int_0^{t_f} P(t)\,\mathrm{d}t \tag{13}$$

The t_f in Equation 13 represents the final time of the definite integral. However, while an integral requires a continuous function, there are only a finite number of data points. Thus, a trapezoidal Riemann sum was used to approximate each integral in the form shown in Equation 14.

$$E \approx \frac{\Delta t}{2} \sum_{n=0}^{t_f - 1} (I(n) \cdot V(n) + I(n+1) \cdot V(i+1)) \tag{14}$$

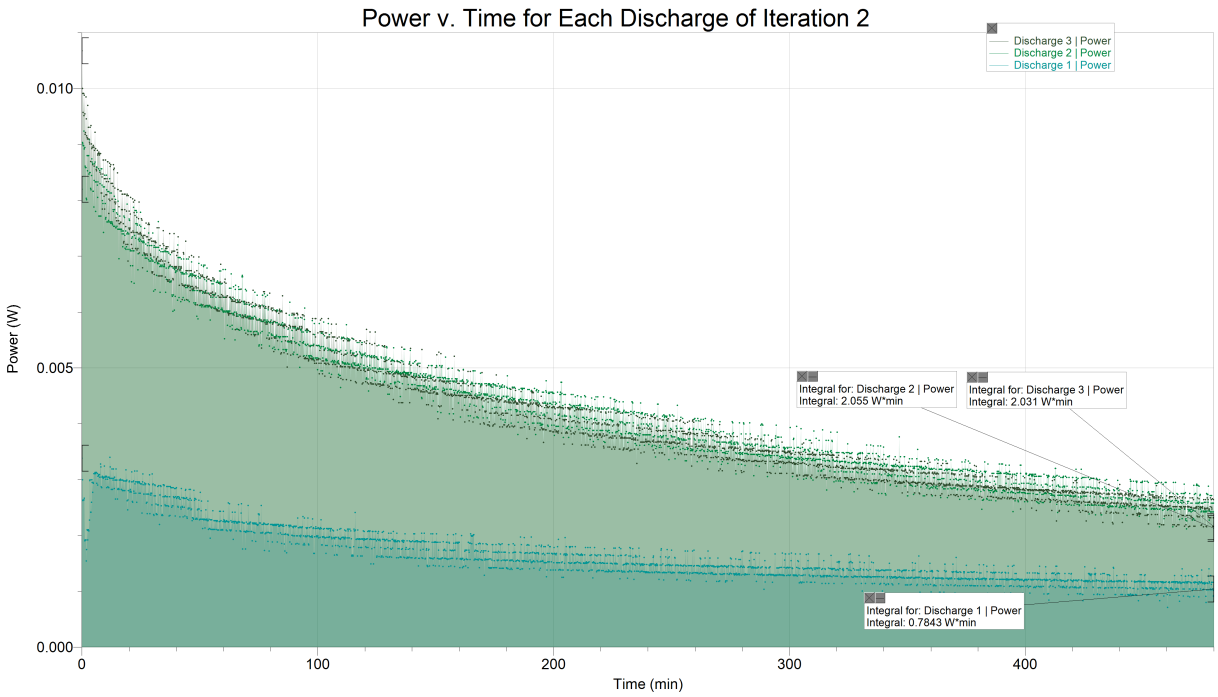


Figure 45: A graph showing the power versus time and its integral for Iteration 2.

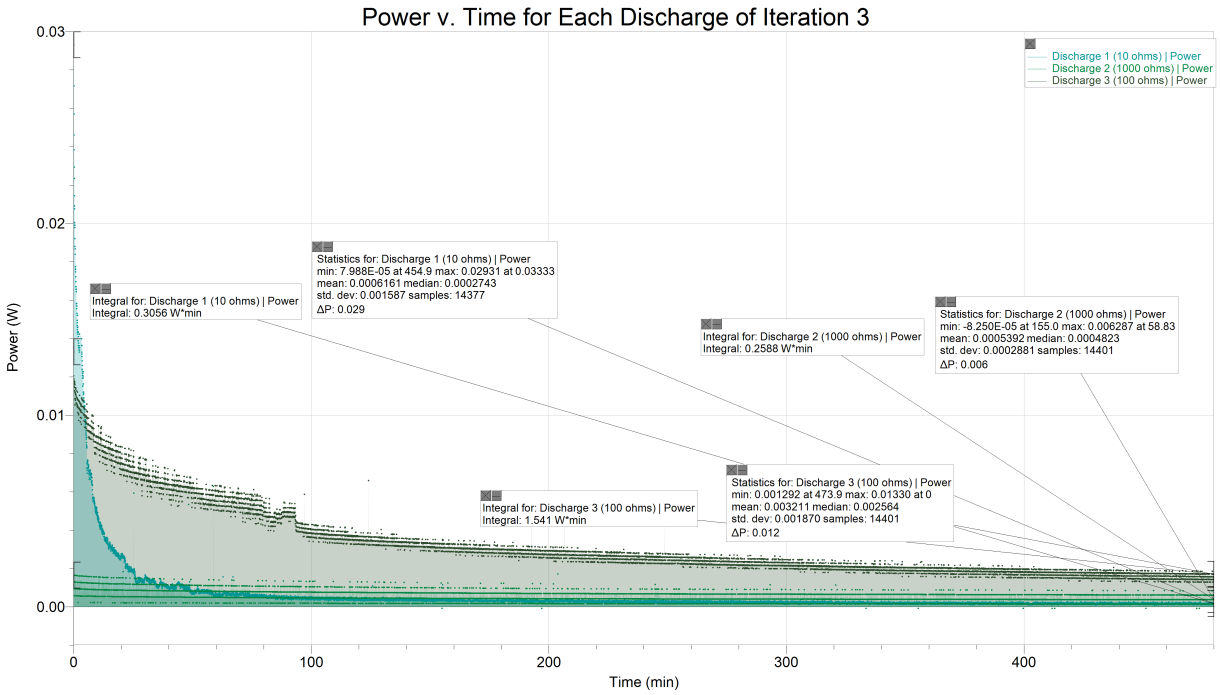


Figure 46: A graph showing the power versus time and its integral for Iteration 3.

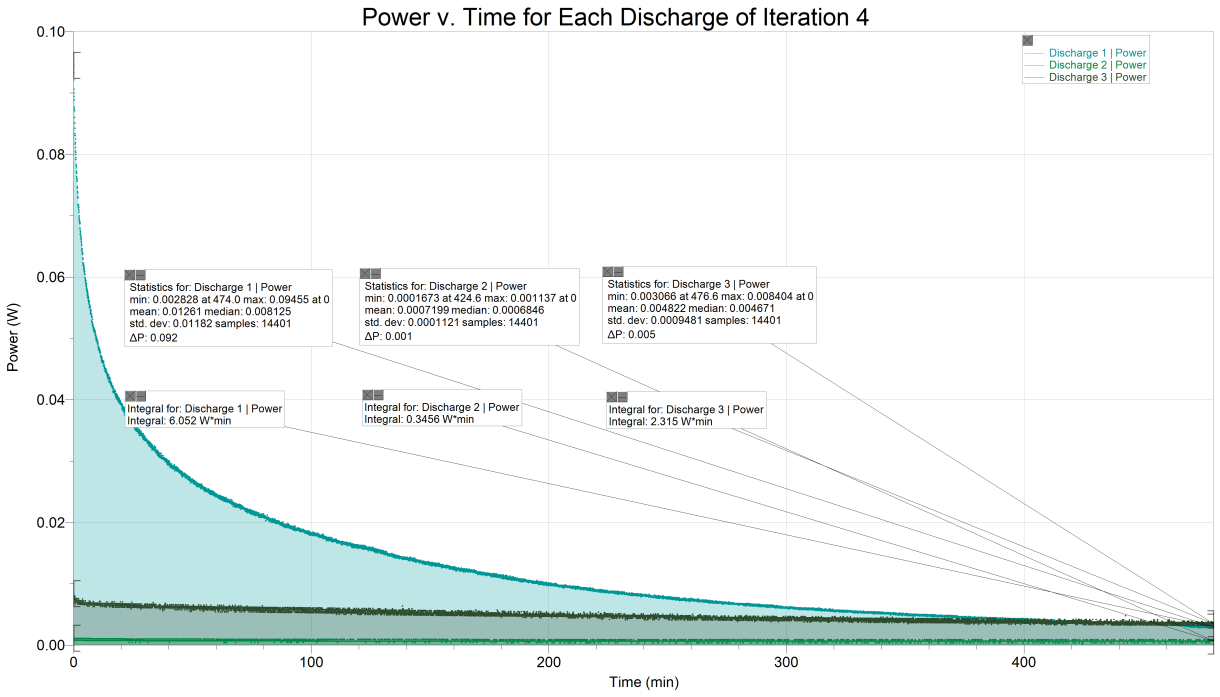


Figure 47: A graph showing the power versus time and its integral for Iteration 4.

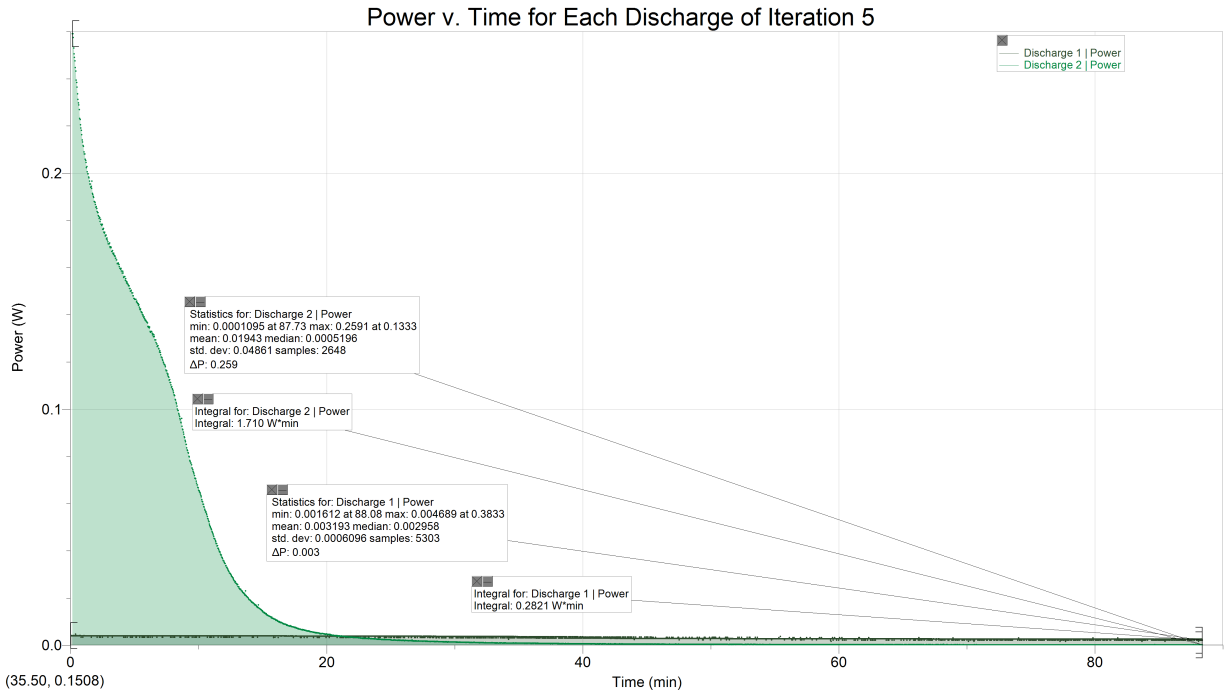


Figure 48: A graph showing the power versus time and its integral for Iteration 5.

The effective charge capacity is a metric of how much charge can the battery transfer within a certain voltage range. Similarly, the effective energy of a battery is determined by how much energy is released within a useful voltage range. A AA battery, like the NiMH battery used for comparison, is only useful when the voltage is greater than 0.9 V on average (Energizer, 2010). Thus, finding the area beneath each curve of current v. time and power v. time while the voltage is greater than 0.9 V would give the effective charge capacity and effective energy, respectively. Thus, Table 1 shows the effective capacity and effective energy for iteration 2. The values are given in the derived units from the graph and in standard SI units.

Table 1: The effective capacity and effective energy of each discharge for Iteration 2.

	Discharge 1	**Discharge 2**	**Discharge 3**
Effective Capacity (A min)	0	0.5647	1.196
Effective Capacity (C)	0	33.88	71.73
Effective Energy (W min)	0	0.5200	1.125
Effective Energy (J)	0	31.20	67.51

Table 2: The effective capacity and effective energy of each discharge for Iteration 3.

	Discharge 1	**Discharge 2**	**Discharge 3**
Effective Capacity (A min)	0	2.818	5.648
Effective Capacity (C)	0	169.1	338.9
Effective Energy (W min)	0	2.316	5.498
Effective Energy (J)	0	138.98	329.9

Table 3: The effective capacity and effective energy of each discharge for Iteration 4.

	Discharge 1	**Discharge 2**	**Discharge 3**
Effective Capacity (A min)	1.009	0.8681	0.02586
Effective Capacity (C)	60.54	52.09	1.552
Effective Energy (W min)	0.09455	0.8073	0.02339
Effective Energy (J)	5.673	48.44	1.404

Table 4: The effective capacity and effective energy of each discharge for Iteration 5.

	Discharge 1	**Discharge 2**
Effective Capacity (A min)	2.093	1.310
Effective Capacity (C)	125.6	78.59
Effective Energy (W min)	0.004071	0.2138
Effective Energy (J)	0.2443	12.83

To allow for effective comparison between the different batteries, each of which was made with different chemistries and showed different charge/discharge graphs, each charge and discharge was normalized by dividing every value in a charge or discharge by the first value. An example of the normalization formula is shown in Equation 15. Due to being defined as a quantity at some time divided by the initial value of that quantity, normalized versions of a quantity do not have any units. Since the

normalized values can be easily compared though the actual values may differ substantially, all of the normalized graphs contain the best curve fit approximations of each graph.

$$V_{\text{norm}} = \frac{V}{V_0} \tag{15}$$

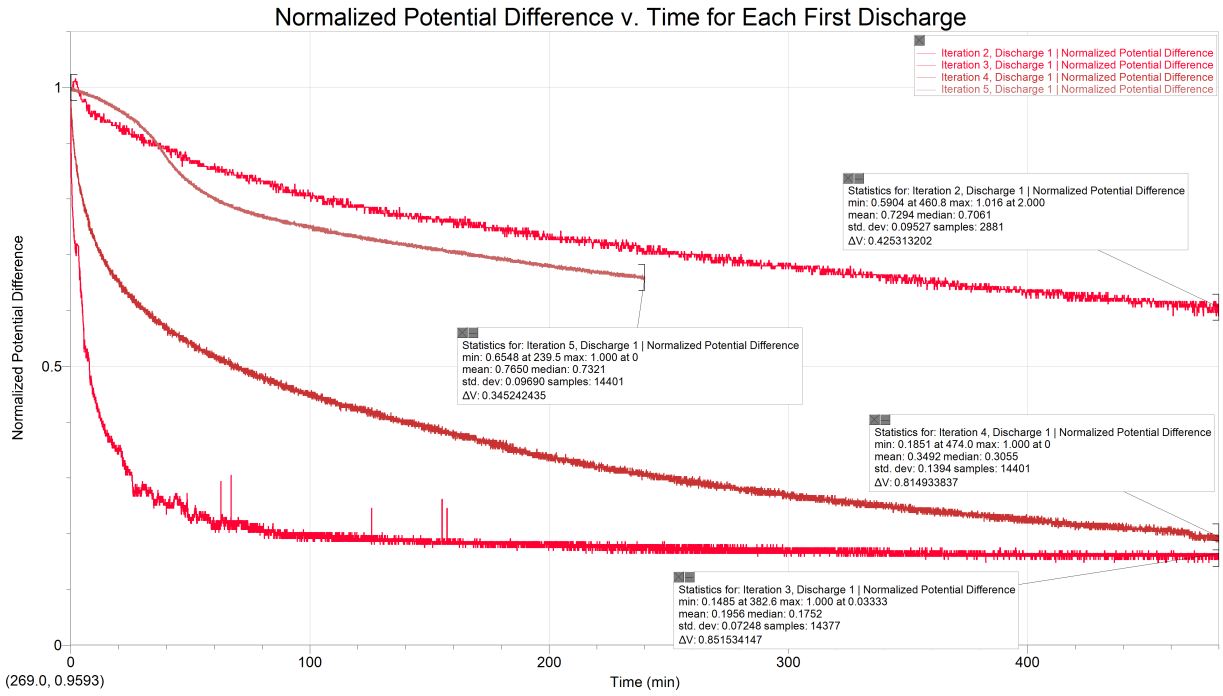


Figure 49: A graph of all of the normalized voltages versus time for the first discharge of each iteration.

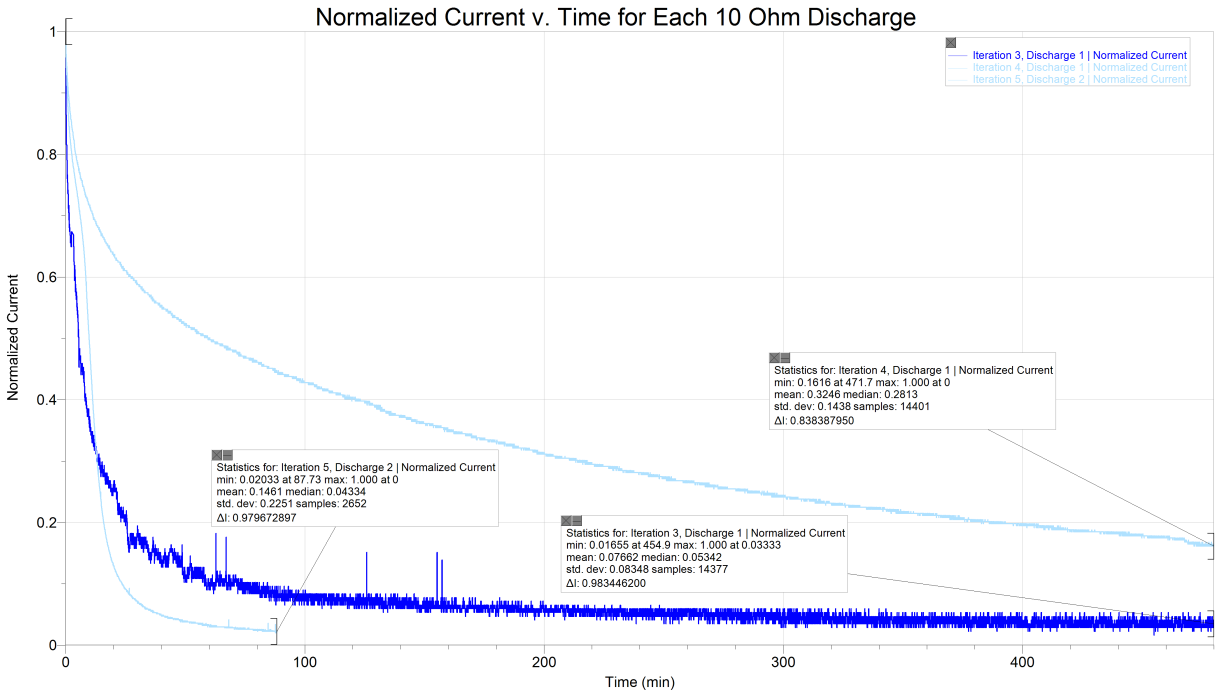


Figure 50: A graph showing the current v. time for all of the discharges done at 10 ohms.

The internal resistance of a battery was calculated from the following equation:

$$V = \varepsilon - Ir \tag{16}$$

In this equation, V represents the terminal voltage, ε is the emf, I is the current, and r is the internal resistance. Rearranging this equation for r gives Equation 17. While the emf of the battery cannot be found during a discharge, the emf at the beginning and end of each discharge were known from the open-circuit voltage that is measured at the beginning of each discharge (as the data collection was started just before the circuit was wired), or from the charging tests, which involved directly measuring the emf. Using those emf values, along with the first or last voltage and current values measured for a discharge, the internal resistance can be determined at the beginning and end of each discharge, using the formula listed in Equation 17.

$$r = \frac{\varepsilon - V}{I} \tag{17}$$

Table 5: A table of the internal resistances at the beginning of each discharge.

Resistance (Ω)	Iteration 2	Iteration 3	Iteration 4	Iteration 5
Discharge 1	78.66	2.406	4.823	75.71
Discharge 2	18.07	56.55	89.87	4.282
Discharge 3	4.146	26.70	2.350	N/A

Table 6: A table of the internal resistances at the end of each discharge.

Resistance (Ω)	Iteration 2	Iteration 3	Iteration 4	Iteration 5
Discharge 1	151.81	150.90	36.05	108.98
Discharge 2	28.32	174.74	188.3	182.3
Discharge 3	45.43	93.30	23.73	N/A

Discussion

While some comparisons have been made here between all the batteries in general, and in comparison to the NiMH battery that was also discharged, the vast majority of comparison within this discussion section was done between successive iterations and within a single iteration. As mentioned before, no data was collected for iteration 1, due to a lack of reliability and standardization in the data.

Figure 20 through Figure 23 describe the discharges of the second battery. The most striking feature of Figure 20 appeared to be the high amount of correlation between discharges 2 and 3. The mean and median values of discharges 2 and 3 lie within 5% of each other, while the change in potential (represented by the Δy on the statistics of each curve), shows greater variation. The curves all seemed to be converging onto a similar value, and an exponential function was found to be a good model of the battery behavior for $t > 30$ min. However, for $t < 30$ min, the battery potential dropped at a steeper rate than could be modeled for an exponential function. The most likely explanation for this is the electrical double layer capacitance explained earlier; ions adsorb onto the activated carbon very quickly, allowing for higher initial discharge currents, but this process cannot sustain for long.

It is therefore likely that the intercalation process that should dominate the discharge after a long time has passed (in this case, most likely when $t > 30$ min) exhibits exponential decay. This can be explained using rate kinetics. The chemical equation governing the oxidation of zinc was shown in Equation 1. This equation is an elementary reaction (as nothing simpler would be possible), and so the reaction order can be determined from the stoichiometric coefficient of 1. The intercalation equation of zinc into graphite was shown in Equation 3, but is shown here again for clarification:

$$\mathrm{Zn}^{2+}(\mathrm{aq}) + x\mathrm{C}(s) + 2e^- \longrightarrow \mathrm{ZnC}_x(s) \tag{18}$$

It can be safely assumed that this Equation 3 is an elementary reaction, as no simpler reaction is possible. As graphite (C) and the zinc-graphite intercalation compound (ZnC_x) are both solids, the reaction must be first order and based on the concentration of zinc ions. The general equation for first order kinetics can be described through the following differential equation:

$$\frac{\mathrm{d}C}{\mathrm{d}t} = -kC \tag{19}$$

$$C = C_0 e^{-kt} \tag{20}$$

In Equation 19, C represents the concentration (in this case, of zinc ions) and k is the rate constant for the specific reaction. The solution to this differential equation is well known and can be seen in Equation 20. However, adding together the anode and cathode equations gives Equation 21, and simplification down into Equation 22 shows that the overall concentration of zinc-ions does not change. Therefore, Equation 20 can only be applied over a localized area, which would most likely be around the anode or cathode. A local concentration gradient could heavily impede electrical conductivity, leading to the similar (but not quite) to exponential decay of voltage visible in Figure 20.

$$\mathrm{Zn}(s) + \mathrm{Zn}^{2+}(s) + x\mathrm{C} + 2e^- \longrightarrow \mathrm{Zn}^{2+} + 2e^- + \mathrm{ZnC}_x \tag{21}$$

$$\mathrm{Zn}(s) + x\mathrm{C} \longrightarrow \mathrm{ZnC}_x \tag{22}$$

Since iteration 2 was discharged 3 times at $100\,\Omega$, the currents measured were all very low; so low, in fact, that the current sensor could barely pick up many of the fluctuations. The very low current led to what looks like horizontal lines on the graphs of Figure 21, Figure 22, and Figure 23, as the current probe reached its maximum precision. The maximum current of iteration 2, as visible in the statistics of

Figure 23, was 0.01049 A at $t = 0$ min during the third discharge. The integrals of the graphs shown in Figure 21, Figure 22, and Figure 23 show the total amount of charge pumped through the battery over the 8 hours. While Figure 23 shows the highest current, the greatest charge capacity can be seen in Figure 22, as the integral of Figure 22 is the greatest, at $3.079\,\mathrm{A\,min} = 184.7\,\mathrm{C} = 51.32$ mAh. A standard AA NiMH battery has a useful charge capacity of 1100 mAh, while the maximum output from the second battery in a single discharge was about 21.43 times less. However, the much lower quality and simpler designs used in this experiment when compared with a commercial battery built to maximize energy density can account for this massive difference in charge capacity.

Figure 24 shows the emf v. time for each charge of the second battery. As mentioned before, the emf is equivalent to the open-circuit voltage, so the circuit was opened with a timer to find the emf on regular intervals. For all 3 charges shown in Figure 24, the maximum emf was reached at the end of the charge, as is to be expected. It can also be seen that the higher the initial emf was, the smaller the overall change in emf was as well. The emf of a battery will approach that of the power supply, similar to how the voltage drop across a capacitor approaches the voltage of a power source. If while charging, the battery is approximated as a capacitor, the following equation for the voltage of a capacitor can be used to estimate the voltage of the battery:

$$V = V_{\max}\left(1 - e^{-\frac{t}{\tau}}\right) \tag{23}$$

In Equation 23, V is the voltage at any given time, $V_{\max}$ is the maximum voltage across the battery (which is equivalent to the cell potential), t is the time, and τ is the time constant for the equation. However, the curve fits of Figure 24 are of the following form:

$$V = Ae^{-Ct} + B \tag{24}$$

However, Equation 24 can be rewritten in the format of Equation 23 if a time offset is inserted into Equation 23, as shown in the following derivation:

$$V = B\left(1 - \left(-\frac{A}{B}\right)e^{-Ct}\right) \tag{25}$$

$$V = B\left(1 - e^{-Ct + \ln\left(-\frac{A}{B}\right)}\right) \tag{26}$$

Substituting $V_{\max}$ for B, $\frac{1}{\tau}$ for C, and $\frac{t_0}{\tau}$ for $\ln\left(-\frac{A}{B}\right)$ creates the following equation:

$$V = V_{\max}\left(1 - e^{-\frac{t-t_0}{\tau}}\right) \tag{27}$$

The only difference between Equation 27 and Equation 24 is the addition of a time offset. However, $V_{\max}$ can be calculated from the equations shown on the graphs through the values of B. Thus, for the second battery, the predicted maximum emfs from the equations are 0.9885 V, 1.056 V, and 1.018 V. However, the statistics from Figure 24 show that the maximum emf of the battery during charge 2 was 1.061 V, which is slightly higher than the predicted value. This shows the limitations of modeling a battery as a capacitor and that the exponential model may not account for variations in the battery rising from the chemical reactions occurring to generate the cell potential.

Similar to the graph of potential difference v. time shown in Figure 20, the graph shown in Figure 26 shows a much lower initial potential during the first discharge than the others. Since the battery was not charged before the first discharge, this is to be expected. Discharge 1 was performed at 10 ohms, so the very steep drop-off in terminal voltage is expected, as the internal resistance rises quicker when the current is higher. It seems that something happened to the third battery around 1.5 hours after discharge 3 began, as there is much greater variation in that area. This variation is most likely caused by some outside disturbance that causes a change in the battery. For example, if some shaking in the environment had caused the cathode and anode to move closer together, the distance ions would need to travel to complete the circuit would decrease, decreasing the internal resistance and thus increasing the terminal voltage. It seems that as the batteries went through dischage-charge cycles, the terminal voltage over time seems to smoothen. This can be seen with the in the second discharge graph and the discharge 3 graph (apart from around $t = 90$ min). The discharge 1 graph seems much more rough, which is likely due to the higher discharge currents and the "break-in" effect mentioned before, where battery performance can improve after a discharge.

Figure 27 shows the current in the third battery during the first discharge. Similar to what was mentioned for the terminal voltage of the first discharge, there was very steep drop-off in current within the first hour of the discharge. More surprising at the time was the very low current value, which the statistics for Figure 27 show to reach a max of 0.05 A. However, as was seen in Figure 26, this is due to the low potential difference that the third battery had during its first discharge.

Figure 28 is almost impossible to interpret due to the tiny currents which bordered on the edge of the precision of the current probe. The current was so low that because of the limit of precision on the current probe, the minimum value for Figure 28 reads a negative value. This is to be expected, because the battery was disharge across a 1000 ohm resistor. With such high resistance, a very low current is expected, leading to the difficult to interpret results of Figure 28.

During discharge 3, as shown in Figure 29, the third battery performed much better, as the integral of Figure 29 was found to be greater than that of Figure 27, even though the external resistance was 10 times larger. From this, it can be inferred that the third battery's performance improved considerably after charging and with a few "break-in" cycles done first. If another 10 ohm discharge had been done for this battery, it is likely that the maximum current and charge capacity would have been much higher.

As mentioned in the procedure, the cathode of Iteration 3 was placed into zinc sulfate solution instead to test the performance obtained from mixing the two electrolytes. This is shown in Figure 30 and Figure 31. By far, the most noticeable effect in both graphs is the extremely steep slope of the curve at the beginning, followed by a somewhat constant value afterward. In Figure 30, the statistics show that the average voltage was 0.08890 V, while Figure 31 shows that the average current was 1.617 mA. Such a low potential and current implies that the energy stored in the modified Iteration 3 battery must have been a small quantity and left the battery very quickly. When there are more available sites for intercalation in the graphite cathode, the potential difference is larger. Thus, from this graph, it can be inferred that the amount of intercalation reactions occuring was very limited. Thus, if the performance of the modified Iteration 3 battery were to be improved, charging the battery for an extended period of time before a discharge test may have produced better results. However, due to the extreme drop in voltage and current that occurred in this battery almost immediately, it was deemed unnecessary to expand research on this battery, as its performance was far poorer than all of the rest of the batteries.

Figure 32 shows the emfs while charging iteration 3. Similar to Figure 24, the battery emf increased over time, approaching the value of the power supply. From the curve fits of Figure 32, the expected maximum emf values are 1.338 V for the first charge and 1.466 V for the second charge. These

values are much higher than the respective values for iteration 2, which most likely results from the switch of electrolyte. However, the exponential curve does not correlate as well with the first charge. A possible cause for the spikes seen on the graph may be a misaligned timing circuit, causing the voltage to be read while the circuit was closed. If that was the case, then those values show the power supply voltage at those times. However, the spikes would be expected to happen more often than twice, with a 15 minute gap in between. The power supply voltage possibility also does not explain the much lower voltage in at the next data point that was collected, unless also caused by a misalignment of the timing circuit.

Figure 33 can be analyzed in a similar fashion to Figure 25, though there seems to be much more of a trend in the charges of iteration 3. The current does start higher and end up lower, though there is considerable variation in the current during the process as well. The current in this case can be used as a metric of the reaction rate: the higher the current, the faster the reverse reactions are happening, causing the battery to recharge. Thus, since the reactions were modeled in Equation 20 to follow exponential decay, in the same way, the current show exponential decay. As there is significant variation within Figure 33, there are likely some other processes which have a significant (if not dominant) effect on the current as time passes.

Figure 34, similarly to Figure 20 and Figure 26, shows the first discharge to have a much lower potential than the other two, for the same reasons as explained before. However, in both Figure 26 and Figure 34, the second discharge started and stayed at a higher potential than the third discharge, while this effect was not visible in Figure 20. While this is most obviously due to the 10 times lower resistance in the third discharge of each battery, another factor could be in the battery chemistry. In the third and fourth iteration of the battery, zinc chloride was used in place of zinc sulfate. The chloride ion is much more reactive than the sulfate ion, so even if no precipitates were to form, the greater amount of side reactions (especially during recharging) could lower the voltage and current of the battery, even only after 2 discharges.

The most amount of charge pumped through any battery can be seen in Figure 35, with the total ouput being an incredible 2.818 Ah. This is almost 55 times better than the best discharge of iteration 2.

However, even more surprising was the max output current of over 1 A, which should not be possible through a 10 ohm resistor with a 1.5 V battery. Using the maximum terminal voltage for discharge from Figure 34 (the maximum voltage and current both occured at $t = 0$ min), Ohm's Law can be used to calculate the external resistance, leading to the following equations:

$$V = IR \tag{28}$$

$$R = \frac{V}{I} = \frac{0.09371\,\text{V}}{1.009\,\text{A}} = 0.09287\,\Omega \tag{29}$$

From Equation 29, it can be seen that this battery must have short-circuited, as $0.092\,87\,\Omega$ is much more likely to be the total resistance of all of the wires. Unfortunately, a battery that is able to short circuit and be completely fine means that the internal resistance of the battery must have been the only load for the battery's current. Assuming that the cell potential was around 1.5 V, the internal resistance would followed this calculation:

$$V = \varepsilon - Ir \tag{30}$$

$$r = \frac{\varepsilon - V}{I} = \frac{1.5\,\text{V} - 0.0937\,\text{V}}{1.009\,\text{A}} = 1.394\,\Omega \tag{31}$$

While this value is quite good, the value given in Equation 31 is still orders of magnitude larger than that of a commercial battery (which is why those batteries should not be short-circuited).

Figure 36, similar to Figure 28, cannot be analyzed effectively, as the current values are so small that the graph is essentialy at a constant value. However, Figure 37 shows much more useful results, as this discharge did not involve a very high or very low resistance. These values are very similar to that of Figure 29, yet the maximum current of Figure 37 was smaller while the charge capacity was higher. This can be explained through the use of much more graphite powder in the construction of iteration 4, causing more intercalation and less adsorption. More adsorption caused the current in Figure 29 to be higher, while more intercalation caused the total amount of charge pumped through the battery in Figure 37 to be higher, as intercalation is a far slower process that results in steadier currents, while adsorption relies on the electrical double layer to produce a capacitative effect.

It is difficult to interpret Figure 38, as the emfs do not follow any one trend. Thus, unlike Figure 24 and Figure 32, no curve fits were done for Figure 38. The first charge shows that sudden spikes

that were visible in Figure 32, yet it is sustained for a little while. Again, this can be put down to misalignment of the timing circuit and the timing of the data collection. Other than those three values, however, the emf in charge 1 seems to steadily increase. However, the emf in charge 2 shows a slightt decrease over time, with the maximum value being the very first emf measured. The most plausible reasons for this are that the battery was already charged to the voltage of the power supply or the current dropped to 0 for one reason or another. The first answer is unlikely, as the other two discharges easily surpass the emf values shown in scharge 2. Thus, the current most likely dropped to close to 0.

That current drop for charge 2 is evident in Figure 39, as the current value went from a max of 0.1062 A down to around 0.02708 A after the first data point. The second data point was recorded 2.5 minutes after the first, so something must have occurred within that time to bring the current close to 0. If a short-circuit had occurred somewhere in the circuit, it is possible that the power supply detected that short-circuit and set the current to almost 0 in order to prevent any damage from occurring via that short circuit. The graps for charges 1 and 2 look more regular, but again, there is no clear exponential trend. There are random spikes and drops in the current, though the overall trend is clear: some sort of decay function, trending towards 0.

The most interesting part of Figure 40 was the inflection point visible on the graph, which is characteristic of many standard battery discharge graphs. In a commercial battery, the potential stays at close to the nominal value for a long time, and only drops at the very end of the battery's lifetime. Something similar can be seen here in both discharges 1; the potential stayed above 1.8 V for 40 minutes before the reaching the inflection point and switching from a battery-like discharge to a capacitor-like discharge (exponential decay).

The timespan of Figure 40, as well as Figure 41 and Figure 42, was greatly limited due to adverse effects while charging the battery. Due to the higher potential at which Iteration 5 was charged, chlorine gas began to form and reached a high enough concentration in the air to become noticeable. To mitigate the adverse effects of chlorine gas, a door was opened to allow fresh air from the outside to come in. The temperature had been kept constant at 18.5 ℃, but allowing air from the outside caused the temperature to drop to 5.3 ℃. Batteries perform worse at lower temperatures due to decreased reaction rates, causing

a steep drop in the potential and current of the battery. As these effects were most noticeable later in the discharges, it was decided to cut the data off at $t = 90$ minutes to ensure the data would be comparable to the other iterations, while the effects from the production of chlorine gas rendered the charging graph unreliable and not fit to use.

Figure 41 is another example where the small current values began to reach the limit of the current probe's precision. As expected from the much smaller timespan and very small current, the charge pumped through the iteration 5 battery was much smaller than many of the values of iterations 2-4. However, Figure 42 shows much higher current values and the inflection point (the "hump") that is only seen in battery discharge graphs. Neverthless, the current through the battery becomes so low as to be negligible after even just 1 hour. It is likely that this is due to the general cooling of the environment caused by the opening of the door. Even in this short time, the fifth iteration battery was able to pump a considerable amount of charge ($2.106\,\mathrm{A\,min} = 126.36\,\mathrm{C} = 35.1$ mAh). Thus, by combining the changes made to iteration 4 and iteration 5 along with other design changes, considerable improvement could be made in future experiments to produce a battery which can compete with many of the current commercial batteries.

Figure 43 and Figure 44 show the voltage and current while discharging an NiMH battery across a 1000 ohm resistor. With such a high resistance (and therefore a very low current), it was not surprising that the NiMH battery voltage showed almost no drop over the 8 hour discharge. However, from Figure 44, the total charge pumped by this battery was $0.4568\,\mathrm{A\,min}$. While the other two 1000 ohm discharge graphs, Figure 28 and Figure 36, did not have quite as high of a value, Figure 36 got quite close with $0.4180\,\mathrm{A\,min}$. Nonetheless, this considerable gap between the 8 hour charge capacities show that the current through the zinc-ion batteries was quite hampered by the much higher internal resistance, which still had a considerable impact even when the external resistance was so large.

The points for discharges 2 and 3 are almost indistinguishable in Figure 45, which can be explained for the same reasons as to why the two discharges were so close in Figure 20 and between Figure 22 and Figure 23. The energy stored by a battery is generally measured in mWh for smaller batteries like those made here. The total energy output from discharge 2 of iteration 2 (and thus,

approximately the energy output from discharge 3) was 2.055 W min = 34.25 mWh. Using the effective charge capacity of a NiMH mentioned before and the voltage of an NiMH battery, the total effective energy of a NiMH battery is as follows:

$$1100 \text{ mAh} \times 1.2 \text{ V} = 1320 \text{ mWh} \tag{32}$$

As can be seen in Equation 32, the energy stored in a single AA battery is far greater than what has been produced here. This does not even account for the effective capacity of the batteries, which will be discussed later. Nonetheless, there is substantial room for improving the quality and design of zinc-ion batteries, as can be seen from these numbers.

The results for Figure 46 are not much better, though there is an interesting extreme spike in power that can be seen at the very beginning of the first discharge. This could be due to a high starting current which rapidly drops off in Figure 27. Surprisingly, the total energy output in discharge 1 and discharge 2 are almost identical, even though discharge 1 has the much higher initial output. However, this is most likely due to the much steadier current and voltage in discharge 2 due to the 1000 ohm resistance. None of the discharges of iteration 3 produced as much power as discharges 2 and 3 of iteration 2.

Due to the short-circuit visible in Figure 35, the energy output of discharge 1 of iteration 4 in Figure 47 is much greater than any of the other discharges. However, due to the extremely low terminal voltage, the actual power was far lower. If the terminal voltage had stayed at a much higher value (close to the cell potential), then the first discharge of iteration 4 would have had a greater total power output than a AA battery. The third discharge of iteration 4 also shows a considerable amount of energy released. With the higher loads, the batteries function much more like the constant voltage source a battery is supposed to be, due to the current being lower in Equation 16. Thus, at higher loads, the zinc-ion battery is able to produce a constant power output. However, that power output is very small; the average power output for discharge 3 in Figure 47 is 4.822 mW. Not much can be done with so little power.

In Figure 48, the power output from discharge 1 also appears to be relatively constant. This can again be attributed to the high load, which leads to small yet somewhat constant current. And again, this

causes the voltage to stay about constant as well due to the effects of Equation 16. Nevertheless, the amount of energy released is quite low, even for discharge 2: $1.710\,\mathrm{W\,min} = 102.6\,\mathrm{J}$, which is about one-third of the energy needed to melt 1 g of ice (Engineering Toolbox, 2019).

The capacity and energy of the various batteries was discussed extensively earlier in this paper, but Tables 1-4 instead have the effective capacity and energy. As mentioned before, this value was found by only using taking the integral of the part of each discharge where the voltage was greater than 0.9 V. To make the quantities easier to understand, each value on the table was expressed in the derived units of the graphs and in standard SI units.

In Table 1, the first discharge of the battery had no useful energy or capacity. This can be attributed to the fact that the battery was not charged before the first discharge, and so the potential difference within the battery was not large enough for the battery to have a useful capacity or energy. The effective capacity and energy for discharge 2 was very close to half of that for discharge 3. Looking at Figure 20, this can be explained because the battery reaches 0.9 V so quickly (within the first half hour) in discharge 2 that what seems to be a slightly longer time spent above 0.9 V for discharge 3 ends up having a massive effect.

Table 2 shows essentially the same effects, with the first discharge not having a high enough voltage for a capacity to be read, and the third discharge having close to double the values of the second discharge. This can be explained for the same reasons as for why these phenomena occured in Table 1. However, the use of zinc chloride in iteration 3 does show some improvements, as the values shown discharges 2 and 3 of Table 2 are much greater (about 5 times larger) than the values shown in Table 1.

The first table showing an effective charge capacity and energy in the first discharge is Table 3. However, since the battery short-circuited during the first discharge, the effective charge capacity was very high while the effective energy was very low. Other than the effective capacity of discharge 1, the values of Table 3 showed much less effective capacity and energy. Thus, it is possible that adding the extra graphite powder in iteration 4 may have contributed to the worse power output and energy output in this iteration of the battery.

Lastly, the values in Table 4 were not impressive, but these batteries were not discharged for the full 8 hours due to the formation of chlorine gas, as mentioned before. Therefore, it is fully possible that the final iteration of the battery could have had much larger values for the effective capacity and energy.

Normalizing the graphs of voltage and current allow for effective comparisons even when the starting values are quite different. Thus, the normalized graphs allow for effective comparisons across all iterations. In Figure 49, all of the normalized voltages v. time for the first discharges of the batteries are shown. While iteration 2 and iteration 3 both started with low voltages in the first discharge, the normalized voltage of iteration 3 dropped much faster than that of iteration 2. In other words, the % loss of voltage was much greater for iteration 3 than for iteration 2 over the same time period. This is most likely due to the zinc chloride, which is more reactive and will produce higher currents, but cannot sustain those currents for as long. The iteration 5 battery is the only one that shows the "hump" that defines a battery. All of the rest of the curves are much more akin to a capacitor, except that an exponential curve fit does fit well, due to the multitude of factors that effect the discharge.

Figure 50 was created instead only at discharges performed with the same external resistance, but there did not seem to be much of a correlation here either. The shape of the graphs, especially that of iteration 3, discharge 1, looks much more like an inverse power function. If the discharge were dominated by second-order kinetics, then an inverse function should model the reaction rate and thus the discharge, but that model is also unable to effectively capture the situation. Due to the inability to create effective curve fits for these graphs, no curve fits were added to remove visual clutter. In the iteration 5 graph shown here, however, almost no "hump" can be seen. Instead, the iteration 5, discharge 2 graph looks like just another discharge graph, though it is much steeper than the others. Again, it is plausible that this was caused by the drop in temperature occurring while this final iteration of the battery was discharging.

The information provided in Table 5 and Table 6 allows for the effective comparison of the internal resistances of the different batteries during the different discharges. As expected, all of the Table 5 values at a specific spot on the table are less than the value at the same location in Table 6. However, there does not seem to be much correlation between the starting and ending internal resistances of the batteries. For example, while the starting internal resistance was lowest for iteration 2 at the

beginning of discharge 3, the lower ending internal resistance for the same iteration was in discharge 2. Some of these values were much closer to a regular battery, such as discharge 1 of iteration 3 and discharge 3 of iteration 4, but there does not seem to be much reason as to why. Discharge 2 of iteration 3 and 4, as well as discharge 1 of iteration 5, were all performed at 1000 ohms, and these were the highest values in those columns for the starting internal resistance. The 10 ohm resistance discharges, such as discharge 1 of iterations 3 and 4, as well as discharge 2 of iteration 5, showed much lower internal resistances. However, this does not explain the decreasing starting internal resistance over time, as visible in the "Iteration 2" column of Table 5. It is instead possible for the iteration 2 battery that repeated discharges at the same resistance allowed for battery to exhibit the "break-in" effect for the first few cycles. To verify this, however, would require many more cycles to see whether the effect would persist.

Analysis of Error

Due to the various and complex interactions occurring within a battery, there are numerous factors which could have had a substantial effect on the results. These factors include the surrounding environment, instrument errors, unwanted side reactions, and material purity, among other things.

While the surrounding environment was kept almost constant for the duration of data collection, the nature of the environment itself may had an effect on the battery. All of the experimentation, up until iteration 5, was conducted within a climate-controlled room at 18.5 ℃ and 35% relative humidity. Even in this environment, however, certain compounds in the air, moisture, and possible movement and shaking of the surroundings may have had an effect on the batteries.

Instrumental errors are an inherent problem of every experiment and can occur due a variety of factors, including: the age of the equipment, defects in production, interactions with the environment, imprecision inherent in the equipment, as well as due to calibration errors. Using newer equipment and checking the equipment for proper function can mitigate the first two errors, but interactions with the environment are often many and hard to control. Keeping the environment consistent, however, can help to mitigate those errors. As the equipment used is not exact, there is only a certain level of precision that can be reached, leading to the limitations seen in all of the current graphs for the 1000 ohm discharges. The voltage probe has a resolution of 3.1 mV, which did not become an issue for this paper (Vernier,

2024). However, the current probe had a resolution of 0.31 mA, which was large enough to become a problem in the analyses of Figure 28, Figure 36, and Figure 41. While no current probes of higher precision were available to be used in this experiment, making sure the equipment used is as precise as possible for the range of values that one will work with can help to mitigate the imprecision of measurement. While automated systems were used to time all of the data collection done in this paper, imprecision in the timing of those systems (such as the Arduino microcontroller or the LabQuest2) could also cause a slight misalignment of current and voltage data. However, as the data was collected for 1 hour or 8 hours, a mistimed reading would not be expected to have a substantial effect. The resistance box used for the majority of the experiment is also not exact, as the resistance values can vary by up to 1. Since these resistance values were not used for any direct calculations, this source of error was not expected to make a significant difference in the results.

In addition to the errors inherent in the instruments used, irreversible side reactions can become a major issue over longer time scales. These reactions happen mostly when charging, and are the primary reason for why batteries lose charge capacity over time. An example comes from the problems found in iteration 5 of this experiment. The formation of chlorine gas while charging takes chlorine out of the solution, lowering the concentration of chlorine and therefore the conductivity of the solution. That is why zinc sulfate is currently the preferred electrolyte, as it is less reactive while still quite effective.

Along with unwanted chemical reactions, dendrites can form on the surface of the zinc while charging. While charging, zinc ions are converted back into zinc metal which essentially electroplate (or form a layer on top of) the existing zinc foil. However, this plating is not uniform, and is instead favored around irregularities on the surface of the foil. As zinc continues to build up around the irregularity, a dendrite forms. It is possible for these dendrites to connect the anode to the cathode and thus cause a short-circuit across the battery, drastically lowering the voltage and current measured. Though possible, this is unlikely, due to the thickness of the filter paper used to create the cathode acts as a separator and ensures that the two electrodes would not come into contact.

One of the largest factors on the efficiency of the batteries is the purity of the materials used. For example, as visible in Figure 4, the activated carbon used in this experiment was described as “food-

grade". While this is not a clear specification, it is obvious that "food-grade" activated carbon will be at a lower purity than "lab-grade" or "industrial-grade" activated carbon. By the same logic, the purity of the zinc foil, zinc sulfate, zinc chloride, graphite powder, and manganese dioxide likely had a substantial effect of the efficiency of the batteries. Those impurities lead to the side reactions and dendrites mentioned earlier, but could also cause degradation of the cathode by preventing effective intercalation and adsorption, among other things.

Conclusion

As renewable sources of energy become more prevalent across the globe, grid energy storage capabilities have needed to expand massively to accommodate sources of energy that do not provide a constant output. While current battery technologies are effective for electronics, cars (gas and electric), and short term energy storage, batteries which can hold the energy for a long time while still being safe and reliable will become key.

The zinc-ion batteries designed and tested in this experiment did not perform as well as even AA battery, but the changes and improvements seen in this experiment between iterations in emf, maximum current, internal resistance, and power output show that there is a lot of potential for zinc-ion batteries. In terms of safety, batteries with zinc sulfate as the electrolyte were safer, but zinc chloride batteries allowed for higher currents at the cost of stability and long-term efficiency. The zinc chloride-based iteration 3 had by far the greatest effective capacity and effective energy. In iteration 5, adding manganese dioxide to the cathode improved performance by stabilizing higher voltages for longer.

Nevertheless, further research is necessary before zinc-ion batteries become a commercially viable product. If this experiment were to be redone with industrial-grade materials, longer discharges, many different electrolytes, and more cathode chemistries, substantial improvements could happen. With these improvements, zinc-ion batteries will help the world convert to renewable energy. These sources will rely on safer, cheaper, long-lasting zinc-ion batteries to create a sustainable world.

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